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The Tragedy of Complexity*

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Abstract

Complexity can create value. At the same time, understanding more complex goods requires more of an agent's attention. We show that equilibrium complexity is generally inefficient when agents face competing demands on their limited attention. Because attention allocation is hump-shaped in complexity, equilibrium complexity is distorted towards intermediate levels: well-understood goods are inefficiently complex, whereas less well-understood goods are oversimplified. We apply our model to financial institutions facing regulatory bodies and CEOs interacting with corporate divisions.

Keywords: complexity; limited attention; attention externality; financial regulation; communication in firms

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1 Introduction

Economic activities differ vastly in their complexity. Some appear overly complex, like the rules set by financial regulators. Others appear oversimplified, for example when the media and politicians discuss policy issues without the required nuance. What these situations have in common is that an economic agent (a household, a financial institution, or a CEO) faces multiple demands on her limited attention. This paper develops a parsimonious model of the economic forces that determine equilibrium complexity in such settings.

The key premise of our model is that complexity can add value. This feature represents a departure from much of the existing literature, which has focused mainly on complexity as a means of obfuscation to increase market power. At the same time, our model captures that when something is more complex—i.e., "not easy to understand or explain"[1]—reaching a certain depth of understanding requires more of the agents' limited attention, which invariably reduces the amount of attention agents can allocate to other goods.

To illustrate, consider a setting in which multiple instructors teach courses to the same cohort of students. Each instructor chooses the complexity level of their course—for example, how nuanced a perspective to provide on the subject matter and how much detail to cover. Instructors do not increase complexity to confuse students; rather, they do so because they genuinely believe students benefit from more detailed coverage. The challenge is that understanding more complex material requires students to allocate more time to the course, yet their total study time remains fixed. We use this example throughout the introduction to illustrate the key mechanisms. The paper's main economic applications, financial regulation and corporate communication, are discussed in Section 5.

Our analysis yields three main results. First, we show that even though complexity can generate value, it is generally inefficient in equilibrium. In contrast to models in which complexity serves merely to obfuscate, we find that complexity can be either too high or too low.

[1] The Britannica Dictionary, https://www.britannica.com/dictionary/complex, last retrieved October 18, 2024.

This inefficiency arises from an *attention externality*. In our example, when instructors choose course complexity, they aim to increase student attention to their subject, thereby enhancing understanding of the material. However, they fail to internalize that attention is a common resource: drawing more attention to one course necessarily reduces the time students can devote to others. Depending on how students reallocate attention in response to changes in complexity, some courses end up oversimplified while others become overly complicated.

Second, we characterize how attention is reallocated in response to changes in a good's complexity. We show that the amount of attention allocated to a good—which we call the *attention curve*—is hump-shaped in the good's complexity: goods of intermediate complexity attract the most attention. The intuition is that very simple goods can be understood with little attention, while consumers essentially give up on extremely complex goods since they know they cannot understand them anyway. Concretely, students pay most attention to courses of intermediate complexity, devoting little time to trivially simple courses or hopelessly complex ones. In equilibrium, this hump shape leads to a distortion toward intermediate complexity. Consequently, goods that would be well understood under the social planner's solution are made overly complex, whereas goods that would be less well understood are oversimplified. In this setting, courses that students could master with reasonable effort end up being made unnecessarily complicated by instructors, while genuinely difficult material is presented in an oversimplified manner.

Third, we show that the direction of the complexity distortion depends on the consumer's attention capacity. When attention capacity is high, consumers understand goods well in equilibrium, and suppliers attract attention by making goods overly complex. When capacity is limited, goods are poorly understood, and suppliers attract attention through oversimplification. In our example, students with substantial study time (such as PhD students) face inefficiently complex courses, while students with limited study time (such as part-time MBA students) are taught oversimplified material.

Our model formalizes these insights. A consumer with limited attention consumes goods from multiple suppliers and decides how much attention to allocate to each good given their complexities. We model limited attention by assuming that the consumer has a fixed time budget to allocate across all goods. Suppliers have market power, allowing them to extract a share of the value generated by their goods. In our example, this corresponds to instructors who care about student understanding of their subject. The key choice variable for suppliers is the complexity of their goods, which they set non-cooperatively.

Key to our model is the assumption that more complex goods require more attention to achieve the same *depth of understanding*. Specifically, when the complexity of a good doubles, it takes the consumer twice as much time to achieve the same depth of understanding. This assumption creates an intuitive trade-off: while a more complex good is potentially more valuable, the consumer must devote more attention to reach the same depth of understanding.

The value generated by good i consists of two components. First, it depends directly on the good's complexity. Holding the consumer's understanding fixed, a more complex good can be more valuable due to additional features or customization. Second, holding a good's complexity fixed, value increases with the time the consumer devotes to understanding the good. Therefore, as in Becker (1965), the consumer's time acts as an input to the value of goods. In particular, a deeper understanding makes the good more valuable to the consumer.

When choosing the complexity of their goods, suppliers internalize that consumers respond by adjusting their attention allocation. Because suppliers have market power and extract a fraction of the value their goods generate, they have an incentive to distort complexity to increase attention paid to their good. However, suppliers do not internalize that attention is a shared resource—drawing more attention to their good reduces the value of other goods, generating an *attention externality*.

Our analysis reveals that the consumer's attention allocation is hump-shaped, with goods of intermediate complexity attracting the most attention. This arises because the payoff from

understanding is bounded: simple goods require little attention to understand, while exceptionally complex goods remain poorly understood even with substantial attention, leading consumers to give up.

Because of this hump shape, relatively simple goods lie to the left of the peak of the attention curve, where attention is increasing in complexity. This gives suppliers an incentive to make these goods overly complex. Since simple goods are relatively well understood, it is well-understood goods that are (perhaps counterintuitively) too complex in equilibrium. In contrast, relatively complex goods lie to the right of the peak, where attention is decreasing in complexity. Suppliers then have an incentive to oversimplify these goods, so that less well-understood goods are "dumbed down" in equilibrium. Irrespective of whether the distortion is toward too much or too little complexity, the shadow price of attention exceeds that in the social planner's solution—the consumer is "pressed for time."

Our analysis predicts where complexity distortions will be most severe. Financial institutions, with large attention budgets and strong incentives to understand regulations, face overly complex financial regulation—a frequent industry complaint. Households with limited attention face the opposite problem: media and politicians oversimplify complicated policy issues. Corporate divisions display both distortions simultaneously: they present simple problems with excessive complexity while oversimplifying difficult ones. In the corporate context, CEOs can eliminate the externality by committing to fixed time allocations per division.

Addressing these distortions is difficult because attention typically lacks a price mechanism—and in many cases (e.g., regulation, communication), the goods themselves are not priced. Market-based solutions exist when goods are priced: competition between suppliers can help, as can requiring suppliers to offer menus of complexity levels, though the latter requires regulation to prevent deviations. An alternative is for consumers to commit ex ante to fixed time allocations across goods, eliminating attention reallocation and the associated externality. However, this demands considerable sophistication. Effective regulation of complexity faces

inherent measurement challenges: assessing complexity levels and their context-dependent value effects is difficult in practice. Our analysis therefore highlights the value of institutional design that incorporates attention constraints from the outset, rather than relying on ex post corrections.

Related literature. By viewing time as an input to the value of consumption goods, our approach to modeling complexity builds on the classic work of Becker (1965). We extend this framework by introducing complexity choice. The choice of complexity affects the value of the good directly, but also changes how the consumer transforms her time into understanding the good. By assuming a limited time budget for the consumer, our framework captures that complexity is inherently tied to bounded rationality (Brunnermeier and Oehmke, 2009) and inattention (Gabaix, 2019). The constraint on the consumer's time serves a role similar to information processing constraints in models of rational inattention (see, e.g., Sims 1998, 2003). The interaction of complexity choice with the consumer's attention allocation differentiates our work from models of quality choice, as analyzed by a literature going back to Spence (1975). Quality increases the value of a good without requiring additional consumer attention, whereas complexity creates potential value that can only be realized when the consumer devotes time to understanding the good. Consistent with our findings, Gonçalves (2024) finds that time spent on a task is non-monotonic in complexity but does not investigate complexity choice, which is the main focus of this paper.

Our paper contributes to the literatures on complexity and competition for attention. The existing literature on complexity has predominantly focused on how firms can use complexity to obfuscate a product's price or quality (see, e.g., Carlin, 2009; Carlin and Manso, 2010; Ellison and Wolitzky, 2012; Piccione and Spiegler, 2012; Spiegler, 2016). A common feature of these models is that complexity is inherently harmful: firms use it to increase market power. In contrast, our model holds market power fixed, highlighting an attention externality that exists even when complexity is not anticompetitive. Moreover, our model allows for a positive,

value-creating role of complexity, generating the novel prediction that equilibrium complexity can be either too high or too low.

The literature on competition for attention has focused mainly on how limited consumer attention affects advertising levels and product market competition. Like our paper, this literature highlights externalities that arise when attention is a common resource. However, this literature has emphasized mechanisms for manipulating consumer attention to reduce price competition, a channel absent in our model. Anderson and de Palma (2012) shows that competition for limited attention generates links across otherwise independent markets, as more attention in one market increases competition there but reduces attention and competition elsewhere. De Clippel, Eliaz and Rozen (2014) show that limited consumer attention introduces an additional dimension of competition across markets when consumers can search for a competitor's price only in some markets. Bordalo, Gennaioli and Shleifer (2016) demonstrate that consumer attention can be drawn to either price or quality, resulting in equilibria that are price- or quality-salient. In Eliaz and Spiegler (2011a,b), sellers seek to influence the options considered by consumers with limited attention and may use loss-making goods as attention grabbers. Hefti (2018) and Hefti and Liu (2020) investigate how limited attention affects pricing and advertising in models of horizontal product differentiation. Johnen and Leung (2022) show that firms disclose excessively detailed information to distract consumers from making price comparisons.

We also connect to several applied literatures. As applications of our model, we highlight financial regulation and communication within firms. As applications of our model, we highlight financial regulation and communication within firms. In the context of financial regulation, Hakenes and Schnabel (2014) develop a model of regulatory capture by sophistication. Asriyan, Foarta and Vanasco (2023) show that regulatory complexity can arise in situations in which the median voter must approve new regulation. In the context of communication within firms, Dessein, Galeotti and Santos (2016) study optimal organizational focus and com-

munication structures in the face of limited attention. Thakor and Merton (2023) study the interaction between product complexity, transparency, and trust in firms.

Finally, complexity has been studied in the context of financial securities. Carlin, Kogan and Lowery (2013) provide experimental evidence that complexity affects volatility, liquidity, and trading efficiency. Célérier and Vallée (2017) and Ghent, Torous and Valkanov (2019) study complexity, promised yields, and realized returns in structured financial products. Ganglmair and Wardlaw (2017) investigate complexity in private loan agreements. Basak and Buffa (2017) analyze how more complex financial models can generate operational risk.

2 Model Setup

We consider an economy with N suppliers (he), who each supply a unit good and choose its complexity.[2] A single representative consumer (she) consumes all N goods. There is one supplier per good $i \in \{1, ..., N\}$. Each supplier has market power. Specifically, we assume that each supplier extracts a share of the value generated by his good. The supplier's payoff is $\theta_i \cdot v_i$, and the payoff to the consumer is $(1 - \theta_i) \cdot v_i$, where $\theta_i \in (0, 1)$. For expositional clarity, we adopt this reduced-form payoff structure in the main text. We provide a more detailed discussion of this specification and other modeling assumptions in Section 4. In Appendix B, we show that our main results extend to a setting with endogenous prices and continuous quantity choice.

One interpretation of these payoffs is that v_i is the value of the good, which is split between the supplier and the consumer according to the weights θ_i (supplier) and $1 - \theta_i$ (consumer). This is the interpretation we use in the text below. An alternative interpretation is that both the consumer and supplier care about the value of the good but assign different utility weights.[3]

[2] Our main model focuses on the symmetric case in which all suppliers choose complexity. However, our main results require only that at least one supplier chooses complexity strategically; the complexity of other goods could be exogenously fixed or chosen by the consumer.

[3] For arbitrary weights that do not add up to one, v_i can always be rescaled to recover the above setup.

The key decision for each supplier is to choose the complexity c_i of the good he designs. While complexity has no direct cost (or benefit) for the supplier, complexity matters because it affects the value of the good. On the one hand, complexity can add value, for example, when it arises as a byproduct of customization that caters to the consumer's specific needs. On the other hand, realizing the full value of a more complex good requires attention from the consumer, who must devote time to understanding it.

The total value of a complex good therefore arises from the combination of its characteristics and the time the consumer allocates to the good. In this respect, our paper builds on classic work on time as an input into the utility derived from market goods pioneered by Becker (1965). Our notion of complexity differs from traditional product quality. Quality increases the value of a good without requiring additional consumer attention. In contrast, complexity creates potential value that can only be realized when the consumer devotes time to understanding the good.[4] This distinction is crucial because complexity affects not only the value of the good itself, but also the consumer's attention allocation across all goods.

Formally, we capture these features of complexity by assuming that the value to the consumer of a unit of good i with complexity $c_i \geq 0$, having allocated $t_i \geq 0$ units of time, is given by

$$v_i\left(c_i, d_i\right), \qquad \text{where} \qquad d_i \equiv \frac{t_i}{c_i}. \tag{1}$$

Here, the first argument of $v(\cdot)$ captures the direct effect of complexity on the value of the good. The second argument captures the effect of the consumer's understanding on the value of the good. How well the consumer understands the good depends both on the amount of attention she devotes to the good and the good's complexity. Note that, as the good's complexity increases, reaching a certain level of understanding requires more attention. Specifically, we

[4]For example, a higher-quality smartphone provides more value immediately upon use, whereas a more complex smartphone (with additional features and customization options) requires the consumer to invest time learning how to use these features.

assume that the consumer's depth of understanding is determined by the time spent on the good divided by the good's complexity. Therefore, a good that is twice as complex takes twice as much time to understand to the same extent. For example, a contract that is twice as long takes twice as much time to read and understand.

We make the following assumptions about the function v_i.

Assumption 1. Functional form of v_i:

(i) v_i *is additively separable*

$$v_i\left(c_i, d_i\right) = f_i(c_i) + g_i\left(d_i\right), \tag{2}$$

(ii) $g_i(d_i)$ *is twice continuously differentiable and* $g_i'(d_i) > 0$, $g_i''(d_i) < 0$ *for* $d_i > 0$,

(iii) $f_i(c_i)$ *is twice continuously differentiable with* $f_i'(c_i) > 0$ *and* $f_i''(c_i) < 0$ *for* $c_i \in (0, c_i^{max})$, *and* $f_i'(c_i) = 0$ *for* $c_i \geq c_i^{max}$, *where* $c_i^{max} \in (0, \infty]$,

(iv) $f_i(c_i) \geq 0$, $g_i(d_i) \geq 0$ *and both are bounded from above,*

(v) the limits of all above derivatives exist as $c_i \to 0^+$ *and* $d_i \to 0^+$,

(vi) $\sigma_i(d_i) = -\frac{g_i'}{g_i'' d_i}$ *is strictly decreasing in* d_i.

Part (i) states that complexity and the depth of understanding enter the valuation of the good in an additively separable form. In addition to making the model tractable, this assumption has conceptual appeal as a baseline case, in which there are no complementarities between complexity and the depth of understanding (i.e., the benefit of understanding a good a little more is independent of its complexity, $\frac{\partial^2 v_i}{\partial c_i \partial d_i} = 0$). In Appendix C we relax this assumption and show that our main results continue to hold when v_i is multiplicative, so that complexity and understanding are complements ($\frac{\partial^2 v_i}{\partial c_i \partial d_i} > 0$). Part (ii) states that the value of the good increases with the consumer's depth of understanding d_i with diminishing marginal return. Part (iii) implies that *ceteris paribus* complexity raises the value of the good. However, as a good becomes more complex, this direct benefit of complexity exhibits

diminishing marginal returns. At some point, the marginal direct effect of complexity on value could become zero.[5] Part (iv) states that goods have non-negative value even if the consumer pays no attention to them. This assumption simplifies the analysis because it guarantees that all goods are consumed in equilibrium. Part (iv) also states that the value of each good is bounded from above. Specifically, goods have finite value even if they have infinite complexity or are infinitely well understood. Hence, suppliers cannot attain infinite value by increasing the complexity of their good, nor can consumers generate infinite value by understanding a good extremely well. Part (v) simply states that the relevant limits exist. Finally, part (vi) is a technical assumption. As will become clear below, this assumption ensures that the consumer's attention allocation is single-peaked in complexity. Most natural choices for f_i and g_i that satisfy the remaining assumptions also satisfy (vi).[6] In Appendix C we show that our main results continue to hold when we relax (vi).[7]

Given that the consumer consumes all goods, the key decision faced by the consumer is how much attention $t_i \geq 0$ to allocate to each of these goods. In making her decision, the consumer takes the complexity of each good as given, but takes into account that she receives a share $1-\theta_i$ of the value v_i generated by good i.[8] The key constraint faced by the consumer is that her attention is limited. Specifically, the consumer has a fixed amount of time T that she can allocate across the N goods. One interpretation of this limited attention constraint is that it introduces an element of bounded rationality, which makes complexity a meaningful concept, see Brunnermeier and Oehmke (2009).

[5] In principle, the direct effect could even become negative. However, because the supplier would never choose a c_i in this region, the assumption that the marginal direct benefit of complexity is weakly positive is without loss of generality.

[6] Examples of such functional forms for $f(x)$ and $g(x)$ are: $\frac{\alpha \cdot x^{\gamma}}{\beta + x^{\gamma}}$ with $\alpha, \beta > 0$ and $\gamma \in (0, 1]$, and $\alpha \cdot \tanh(\beta \cdot x)$ with $\alpha, \beta > 0$.

[7] In fact, for all results through Proposition 2, it suffices to assume that: (i) v_i is twice continuously differentiable in d_i with $\frac{\partial v_i}{\partial d_i} > 0$ and $\frac{\partial^2 v_i}{\partial d_i^2} < 0$ for $d_i > 0$; (ii) v_i is twice continuously differentiable in c_i with $\frac{\partial v_i}{\partial c_i} > 0$ and $\frac{\partial^2 v_i}{\partial c_i^2} < 0$ for $c_i \in (0, c_i^{\max})$; (iii) $\frac{\partial v_i}{\partial c_i} = 0$ for $c_i \geq c_i^{\max}$; (iv) $\frac{\partial^2 v_i}{\partial c_i \partial d_i} \geq 0$; (v) $v_i(c_i, d_i) \geq 0$ and is bounded from above; and (vi) the limits of all derivatives above exist as $c_i \to 0^+$ and $d_i \to 0^+$.

[8] We discuss the model's timing assumptions in more detail in Section 4.

3 The Tragedy of Complexity

In this section, we present the main conceptual result of our paper: Equilibrium complexity is generally inefficient and can be too high or too low. Given Assumption 1, the consumer receives positive utility from consuming good i even when paying no attention to it. It is therefore always weakly optimal to consume all N goods, thus from hereon we simply assume all goods are consumed. We formally define equilibrium and the planner's choice as follows.

Definition 1. Equilibrium *is given by:*

(i) good complexities c_i chosen simultaneously by each supplier $i \in \{1, ..N\}$ to maximize $\theta_i \cdot v_i$, and

(ii) time allocations t_i for $i \in \{1, ..N\}$ chosen by the consumer, taking all c_i's as given, to maximize $\sum_{i=1}^{N}(1-\theta_i) \cdot v_i$ subject to the attention constraint.

Definition 2. The planner's solution *is given by:*

(i) good complexities c_i for $i \in \{1, ..N\}$ chosen by the social planner to maximize total surplus $\sum_{i=1}^{N} v_i$, and

(ii) time allocations t_i for $i \in \{1, ..N\}$ chosen by the consumer, taking all c_i's as given, to maximize $\sum_{i=1}^{N}(1-\theta_i) \cdot v_i$ subject to the attention constraint.

Note that the planner chooses complexity anticipating privately optimal attention allocation decisions by consumers. The planner cannot allocate attention on the consumer's behalf.

We solve the model by backward induction. We first characterize the consumer's attention allocation problem for given good complexities. We then derive complexity chosen by competing suppliers and contrast them with those chosen by a benevolent social planner.

3.1 The Consumer's Problem

The consumer's maximization problem reduces to choosing the amount of attention she allocates to each good, taking as given the complexity c_i of each good,

$$\max_{t_1,..t_N} \sum_{i=1}^{N} (1-\theta_i) \cdot v_i \left(c_i, \frac{t_i}{c_i} \right), \tag{3}$$

subject to the constraint that total attention paid cannot exceed the attention budget,[9]

$$\sum_{i=1}^{N} t_i \leq T. \tag{4}$$

Note that in the maximization problem we have written out d_i as t_i/c_i to highlight the dependence of the depth of understanding on complexity choice.

Denoting by λ the Lagrange multiplier associated with the attention constraint, attention paid to good i satisfies the first-order condition

$$(1-\theta_i) \cdot \frac{\partial v_i \left(c_i, \frac{t_i(c_1,...,c_N)}{c_i} \right)}{\partial d_i} \cdot \frac{1}{c_i} \leq \lambda. \tag{5}$$

This first-order condition holds with equality when $t_i > 0$, in which case the consumer's marginal payoff from an additional unit of attention allocated to good i equals the shadow price of attention λ.

3.2 Equilibrium Complexity: The Supplier's Problem

We now turn to the supplier's choice of complexity. Supplier i's objective is to maximize its own payoff, which is given by a fraction θ_i of the value generated by good i, v_i. The supplier's

[9] By rewriting this constraint as $\sum_{i=1}^{N} \frac{t_i}{c_i} \cdot c_i \leq T$ (i.e., multiplying and dividing by c_i), we see that one can think of the attention constraint as a standard budget constraint in terms of the depth of consumer's understanding, $d_i = t_i/c_i$. In this interpretation, an increase in good i's complexity raises the price for understanding the good. The consumer's wealth is given by her endowment of time T.

only choice variable is the complexity c_i of his good. In choosing c_i, the supplier acts as a Stackelberg leader: he anticipates that consumer attention $t_i(c_1, \ldots, c_N)$ is a function of his complexity choice c_i. Accordingly, the supplier's objective function is given by

$$\max_{c_i} \theta_i \cdot v_i \left(c_i, \frac{t_i(c_1, \ldots, c_N)}{c_i} \right), \tag{6}$$

with the associated first-order condition

$$\theta_i \cdot \frac{d}{dc_i} v_i \left(c_i, \frac{t_i(c_1, \ldots, c_N)}{c_i} \right) \leq 0, \tag{7}$$

which holds with equality whenever $c_i > 0$.

Unless otherwise noted, we focus on the interior case in which the first-order conditions for both c_i and t_i are binding. Taking the total derivative of the first-order condition (7), it can be rewritten as

$$\frac{\partial v_i \left(c_i, \frac{t_i}{c_i} \right)}{\partial c_i} = \frac{\partial v_i \left(c_i, \frac{t_i}{c_i} \right)}{\partial d_i} \cdot \frac{t_i}{{c_i}^2} - \frac{\partial v_i \left(c_i, \frac{t_i}{c_i} \right)}{\partial d_i} \cdot \frac{1}{c_i} \cdot \frac{\partial t_i}{\partial c_i}. \tag{8}$$

This condition states that, from the supplier's perspective, the optimal level of complexity balances three effects. The left-hand side of Equation (8) is the marginal benefit of additional complexity holding the consumer's depth of understanding fixed. The first term on the right-hand side is the resulting loss in value from a reduction in the depth of understanding, holding the consumer's attention to the good constant. The second term on the right-hand side captures the gain or loss in value from the consumer's reallocation of attention in response to the complexity change. In equilibrium, this first-order condition holds for each supplier i.

The key observation is that suppliers take into account that changing the complexity of their good affects the amount of attention that the consumer will allocate to the good, as indicated by the partial derivative $\frac{\partial t_i}{\partial c_i}$ in the optimality condition (8). Crucially, suppliers perceive the

additional consumer attention they attract as a net gain, even though in aggregate attention is merely reallocated: any additional attention paid to good i would otherwise have gone to goods offered by other suppliers. Because the supplier of good i is diverting attention away from other goods, competing suppliers engage in *attention grabbing*.

Substituting in the consumer's first-order condition (5), we can rewrite the supplier's optimality condition (8) in terms of the shadow price of attention λ.

$$\frac{\partial v_i\left(c_i, \frac{t_i}{c_i}\right)}{\partial c_i} = \frac{\lambda}{1-\theta_i}\left(\frac{t_i}{c_i} - \frac{\partial t_i}{\partial c_i}\right). \tag{9}$$

This expression is useful for comparing the supplier's first-order condition to the social planner's optimality condition, derived in the next section.

3.3 Optimal Complexity: The Social Planner's Problem

We now turn to the social planner's choice of good complexity. The key difference compared to the supplier's maximization problem described above is that the planner takes into account that the consumer allocates attention across *all* N goods. Therefore, the planner internalizes the effect of a change in the complexity of good i not only on the value of good i but also, via the consumer's attention reallocation, on all other goods $j \neq i$.

The planner chooses the complexity of the N goods so that they maximize total surplus,

$$\max_{c_1,\ldots c_N} \sum_{i=1}^{N} v_i\left(c_i, \frac{t_i(c_1,\ldots,c_N)}{c_i}\right). \tag{10}$$

Following the same steps as in the derivation of the supplier's first-order condition (including the assumption that the optimal complexity for each good is positive), the condition for the

optimal complexity of good i, c_i^*, is given by

$$\frac{\partial v_i\left(c_i, \frac{t_i}{c_i}\right)}{\partial c_i} = \frac{\partial v_i\left(c_i, \frac{t_i}{c_i}\right)}{\partial d_i} \cdot \frac{t_i}{c_i^2} - \sum_{j=1}^{N} \frac{\partial v_j\left(c_j, \frac{t_j}{c_j}\right)}{\partial d_j} \cdot \frac{1}{c_j} \cdot \frac{\partial t_j}{\partial c_i}. \tag{11}$$

This optimality condition highlights the difference between the social planner's solution and the suppliers' privately optimal complexity choice characterized by Equation (8). In particular, whereas the supplier of good i only takes into account the change in the valuation of good i that results from the reallocation of attention to or from good i, the planner takes into account the valuation changes resulting from the reallocation of attention across *all* goods, captured by $N-1$ additional partial derivatives $\frac{\partial t_j}{\partial c_i}$ on the right hand side. The supplier's privately optimal complexity choice therefore generally differs from the social planner's solution—the reallocation of attention from other goods to good i represents an externality that is not taken into account by the supplier of good i.

As before, using the consumer's first-order condition (5), we can rewrite the social planner's optimality condition (11) in terms of the shadow price of attention λ, which yields

$$\frac{\partial v_i\left(c_i, \frac{t_i}{c_i}\right)}{\partial c_i} = \frac{\lambda}{1-\theta_i} \cdot \left(\frac{t_i}{c_i} - \frac{\partial t_i}{\partial c_i}\right) - \sum_{j \neq i} \frac{\lambda}{1-\theta_j} \cdot \frac{\partial t_j}{\partial c_i}. \tag{12}$$

The second term on the right hand side captures the attention externality on other goods $j \neq i$ that is neglected by the supplier of good i.

3.4 The Complexity Externality

A particularly simple case arises when the share of value going to the suppliers is equal across goods, $\theta_i = \theta$. In this case, the social planner's optimality condition reduces to

$$\frac{\partial v_i\left(c_i, \frac{t_i}{c_i}\right)}{\partial c_i} = \frac{\lambda}{1-\theta} \cdot \left(\frac{t_i}{c_i} - \sum_{j=1}^{N} \frac{\partial t_j}{\partial c_i}\right) = \frac{\lambda}{1-\theta} \cdot \frac{t_i}{c_i}, \tag{13}$$

where the second equality makes use of the fact that, when viewed across all goods, attention is merely reallocated (i.e., $\sum_{j=1}^{N} t_j = T$ implies that $\sum_{j=1}^{N} \frac{\partial t_j}{\partial c_i} = 0$).

The comparison between the supplier's and the social planner's first-order conditions (given by equations (9) and (13), respectively) reveals that there is an externality in complexity choice. The supplier of good i has an incentive to deviate from the socially optimal complexity c_i^* whenever the attention grabbing effect is nonzero at c_i^*, i.e., $\frac{\partial t_i}{\partial c_i}\big|_{c_i=c_i^*} \neq 0$. If this derivative is positive, he has an incentive to raise complexity above c_i^*; if negative, he prefers complexity below c_i^*. In both cases, the distortion is driven by the desire to divert the consumer's attention away from other goods.

Therefore, the crucial step in determining the direction of the externality is signing the direction of the attention grabbing effect, $\frac{\partial t_i}{\partial c_i}$. To do so, it is useful to introduce some additional notation. First, at times it will be useful to write attention as a function of the good's own complexity and the shadow price of attention λ. We denote this function by $\tilde{t}_i(c_i, \lambda)$. Second, note that we can rewrite the value of good i in terms of attention t_i instead of the depth of understanding $d_i = t_i/c_i$. We denote this function by $\tilde{v}(c_i, t_i)$. We then obtain

Lemma 1. Attention Grabbing: Equivalence Results. *For any given vector of complexities $(c_1, \ldots, c_N)$, the attention grabbing effect $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i}$ has the same sign as*

(i) the effect of good i's complexity on the shadow cost of attention, $\frac{\partial \lambda(c_i, c_{-i})}{\partial c_i}$;

(ii) the attention grabbing effect when keeping fixed the shadow cost of attention λ, $\frac{\partial \tilde{t}_i(c_i, \lambda)}{\partial c_i}\big|_{\lambda}$;

(iii) the complementarity (or substitutability) of attention and complexity, $\frac{\partial^2 \tilde{v}(c_i, t_i)}{\partial c_i \partial t_i}$.

Lemma 1 offers three equivalent characterizations of the attention externality. Part (i) says that a supplier gains from increasing complexity above the efficient level when doing so raises the shadow price of attention at c_i^*. Part (ii) gives an equivalent condition in terms of the direct response of attention: the supplier has the same incentive if attention paid to his good rises with complexity, even when λ is held fixed. Part (iii) translates this into the

language of complementarity: the incentive to overcomplicate arises precisely when attention and complexity are complements at c_i^*. In contrast, when they are substitutes, the supplier instead has an incentive to oversimplify.

The result of generic inefficiency, most easily seen in the symmetric case $\theta_i = \theta$, extends to the general case:

Proposition 1. The Complexity Externality. *Starting from the social planner's solution* $(c_1^*, .., c_N^*)$*, the supplier of good* i

(i) has an incentive to increase complexity c_i *if* $\frac{\partial t_i}{\partial c_i}\big|_{c_i=c_i^*} > 0$*;*

(ii) has an incentive to decrease complexity c_i *if* $\frac{\partial t_i}{\partial c_i}\big|_{c_i=c_i^*} < 0$*;*

(iii) has no incentive to change complexity c_i *if* $\frac{\partial t_i}{\partial c_i}\big|_{c_i=c_i^*} = 0$*.*

Proposition 1 states that the complexity externality has the same sign as the attention grabbing effect. Locally, suppliers have an incentive to increase the complexity of their good beyond the optimal level if consumers respond by increasing the amount of attention they allocate to the good. In contrast, when an increase in complexity induces consumers to decrease the amount of attention they allocate to the good, suppliers have a local incentive to reduce the complexity of their good below the socially optimal level.

While complexity may be distorted upward or downward in equilibrium, irrespective of the direction of the equilibrium distortion, the supplier's objective is to divert attention away from other goods. This unambiguously raises the equilibrium shadow price of attention λ.

Proposition 2. The Consumer is Pressed for Time. *Suppose that an equilibrium exists and the complexity of at least one good differs from the social planner's choice. Then the equilibrium shadow price of attention* λ^e *strictly exceeds the shadow price of attention* λ^* *under the social planner's solution,* $\lambda^e > \lambda^*$*.*

The difference between equilibrium complexity and the social planner's solution has parallels with the classic tragedy of the commons. Like grass on a common grazing meadow,

attention is a shared resource. However, in contrast to the classic tragedy of the commons, attention grabbing can manifest itself in too much or too little complexity, depending on whether "overcomplicating" or "dumbing down" leads to an increase in consumer attention paid to a particular good. Yet, whereas the complexity externality can go either way, the scarce resource—the consumer's attention—is always overused, irrespective of the direction of the externality. Specifically, competition for the consumer's attention implies that the shadow price of attention is higher in equilibrium than it would be under the social planner's solution, $\lambda^e \geq \lambda^*$, with strict inequality whenever $c_i^e \neq c_i^*$ for at least one good. Intuitively, when suppliers compete for attention, the consumer feels more pressed for time than under the social planner's solution, irrespective of whether goods are too complex or too simple in equilibrium.

Thus, the conventional tragedy-of-commons intuition applies to attention—the fixed-supply resource shared across all goods. Our contribution is to show that this tragedy of commons in consumer attention generates equilibrium complexity that is generically inefficient and can be too high or too low—the *tragedy of complexity*.

3.5 The Attention Curve

As shown in Proposition 1, the decision to distort complexity depends on the direction of the consumer's attention reallocation in response to a change in complexity. The consumer's response depends on the attention curve $t_i(c_i, c_{-i})$, which captures how much attention the consumer allocates to good i if its complexity were c_i, keeping the complexity of all other goods c_{-i} fixed at their current levels.

We now characterize the shape of the attention curve. Given Assumption 1, we can determine the sign of the complexity externality $\frac{\partial t_i}{\partial c_i}$ by differentiating the consumer's first-order condition (5). Specifically, holding fixed λ and taking the derivative with respect to c_i, for interior $c_i > 0$ we obtain

$$\left.\frac{\partial \tilde{t}_i(c_i, \lambda)}{\partial c_i}\right|_{\lambda} = \frac{g_i'}{g_i''} + d_i. \tag{14}$$

We know from Lemma 1 that $\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i}\big|_\lambda$ has the same sign as $\frac{\partial t_i}{\partial c_i}$. Recalling that $\sigma_i(d_i) = -\frac{g_i'}{g_i'' d_i}$, by Assumption 1 we then obtain:

Lemma 2. *$\frac{\partial t_i}{\partial c_i} > 0$ if and only if $\sigma_i(d_i) < 1$.*

Lemma 2 can be interpreted in terms of the elasticity of demand for understanding a good. When the supplier of good i increases complexity, from the consumer's perspective this raises the price of understanding the good. When the demand for understanding is inelastic (i.e., the own-price elasticity is smaller than one, $\eta_i = \left|\frac{\partial d_i}{\partial c_i}\frac{c_i}{d_i}\right| < 1$), an increase in complexity – the price of a unit of understanding – reduces the demand for understanding the good by less than one-for-one. This less than one-for-one reduction requires that the consumer increases the time spent on the good, so that $\frac{\partial t_i}{\partial c_i} > 0$. The condition in Lemma 2 ensures that $\eta_i < 1$.

To characterize the shape of the attention curve $t_i(c_i, c_{-i})$, we first show that, if an individual supplier increases the complexity of its good, holding fixed the complexity of all other goods, the depth of understanding for this good unambiguously decreases. This means that, even if the consumer reacts by allocating more time to the good, the reallocation of attention only partially offsets the decrease in the depth of understanding due to higher complexity.

Lemma 3. *The depth of understanding d_i resulting from the consumer's attention allocation is strictly decreasing in the good's complexity c_i whenever $d_i > 0$.*

Recall that the payoff difference between understanding a good very well and not understanding it at all is finite (i.e., g is bounded from above and below). Similarly, the payoff difference between a very complex good and a very simple good is also finite (i.e., f is bounded from above and below). Under these relatively weak boundedness assumptions, the attention grabbing effect $\frac{\partial t_i}{\partial c_i}$ must reverse sign at least once, changing from positive to negative as complexity increases. Regularity condition (vi) in Assumption 1 ensures that this sign reversal occurs exactly once, yielding a hump-shaped attention curve.

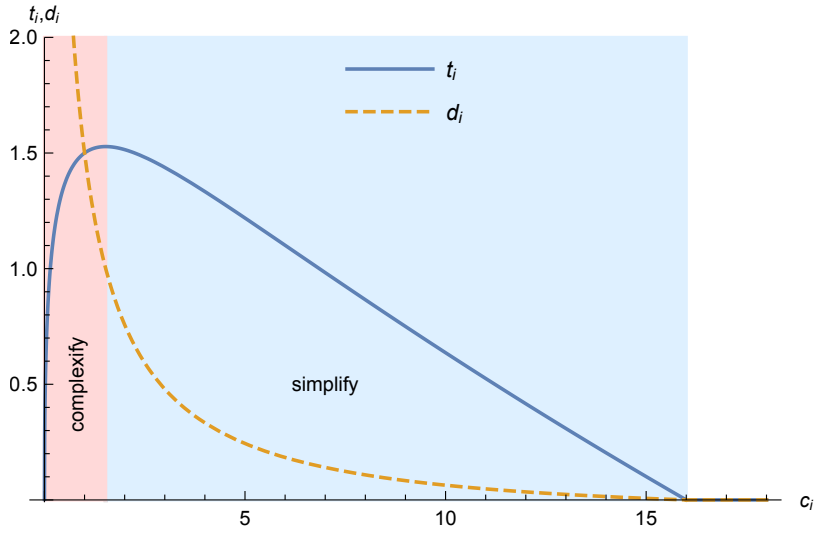
Proposition 3. Single Peaked Attention Curve. *The attention curve $t_i(c_i, c_{-i})$ is single-peaked in c_i,i.e., there exist unique cutoffs $0 < \bar{c}_i < \bar{\bar{c}}_i \leq \infty$ such that*

(i) $\frac{\partial t_i(c_i,c_{-i})}{\partial c_i} > 0$ *for* $c_i < \bar{c}_i$

(ii) $\frac{\partial t_i(c_i,c_{-i})}{\partial c_i} < 0$ *for* $c_i \in (\bar{c}_i, \bar{\bar{c}}_i)$

except for the degenerate case in which $t_i(c_i) = 0$ for all c_i.

Figure 1: **Attention paid to a good as a function of its complexity**



This figure illustrates the consumer's attention choice $t_i(c_i, c_{-i})$ and the resulting depth of understanding $d_i(c_i, c_{-i})$ as a function of the complexity of good i c_i, holding fixed the level of other goods' complexities c_{-i}. Attention allocation is hump-shaped: Initially, $t_i(c_i)$ is increasing, then decreasing, and at some point the consumer chooses to pay no attention to good i. Functional form defined in Proposition 5 with parameters $N = 2$, $i = 1$, $\delta_1 = \delta_2 = 1$, $\alpha_1 = \alpha_2 = 1$, $\theta_1 = \theta_2 = 0.5$, $c_2 = 1$.

Figure 1 illustrates Proposition 3 by plotting the attention curve $t_i(c_i, c_{-i})$ as a function of the complexity of good i for a particular functional form for v_i,holding the complexity of all other goods fixed. Consumer attention follows a hump-shape: For low levels of complexity (up to $\bar{c}$), the curve is upward sloping. In this region, the consumer reacts to an increase in the complexity of good i by increasing attention paid to the good ($\frac{\partial t_i}{\partial c_i} > 0$). Therefore, the supplier of good i has an incentive to increase the complexity of the good. For higher levels of

complexity (from $\bar{c}$ to $\bar{\bar{c}}$), there is a downward-sloping segment. In this region, the direction of the externality reverses: An increase in the complexity of good i leads to a reduction in attention paid to good i ($\frac{\partial t_i}{\partial c_i} < 0$). The supplier of good i then has an incentive to decrease the complexity of the good. Finally, for the specific functional form in this example, above some critical level of complexity ($c_i > \bar{\bar{c}}_i$) there is a region in which $t_i(c_i, c_{-i}) = 0$. In this region the consumer pays no attention to good i even though she still consumes the good. The consumer essentially gives up on learning about the good. Even with significant attention allocated to the good, the consumer would not understand the good well, so that it is better for the consumer to focus her attention on other goods.[10] In this region, the supplier does not have a local incentive to distort complexity.[11]

The hump-shape illustrated in Figure 1 implies that the supplier of good i has an incentive to make goods that are relatively simple too complex and goods that are relatively complex too simple. Of course, whether a good is relatively simple or complex (i.e., whether it lies on the upward-sloping or downward-sloping segment of the attention curve) is an equilibrium outcome that depends on all the parameters of the model. Combining Propositions 1 and 3 with Lemma 2, we arrive at the following proposition which characterizes which goods are most likely to be too simple or too complex:

Proposition 4. Complexity and the Depth of Understanding. *If the consumer pays any attention to good i ($t_i > 0$), there exists a unique $\bar{d}_i \in (0, \infty)$ defined by $\sigma_i(\bar{d}_i) = 1$, such that the supplier has an incentive to*

(i) overcomplicate goods that are relatively well understood in the social planner's solution $d_i > \bar{d}_i$;

(ii) oversimplify goods that are not well understood in the social planner's solution $d_i < \bar{d}_i$.

[10] Examples of this phenomenon include terms and conditions for online purchases and lengthy legal texts, which are classic instances of information overload (Brunnermeier and Oehmke, 2009).

[11] However, the supplier might have an incentive to reduce complexity a larger amount to $c_i < \bar{\bar{c}}_i$ so that the consumer pays attention to the good.

Proposition 4 characterizes which goods are too complex and which are too simple relative to the social planner's solution. Because the consumer's depth of understanding d_i is decreasing in complexity (see Figure 1), goods that are well understood in the social planner's solution lie on the upward-sloping part of the attention curve $t_i(c_i)$. Therefore, suppliers overcomplicate these goods. Conversely, goods that are less well understood in the social planner's solution are located on the downward-sloping part of the attention curve, giving the supplier an incentive to make these goods too simple.

3.6 Symmetric Equilibrium

In this section, we explicitly characterize equilibrium complexity for a specific functional form that fulfills Assumption 1. Given this functional form, we can solve for the equilibrium first-order conditions and characterize the resulting equilibrium in closed form.

Proposition 5. Equilibrium Distortion and the Complexity Paradox. *Assume that* $v_i = f(c_i) + g(d_i)$ *is symmetric across goods with*

$$f(c_i) = \begin{cases} \alpha \cdot c_i - c_i^2 & c_i \leq \frac{\alpha}{2} \\ \frac{\alpha^2}{4} & c_i > \frac{\alpha}{2} \end{cases} \tag{15}$$

$$g\left(d_i\right) = \delta \cdot \frac{d_i}{1 + d_i}, \tag{16}$$

where the parameter $\alpha > 0$ *captures the direct benefit of complexity while* $\delta > 0$ *captures the importance of understanding the good.*

When the benefit of understanding a good δ *is not too large, there exists a symmetric equilibrium in which all goods receive the same amount of attention* $\frac{T}{N}$*. The goods are too complex compared to the social planner's choice if and only if they are well understood in equilibrium*

$$d = \frac{T/N}{c} > 1. \tag{17}$$

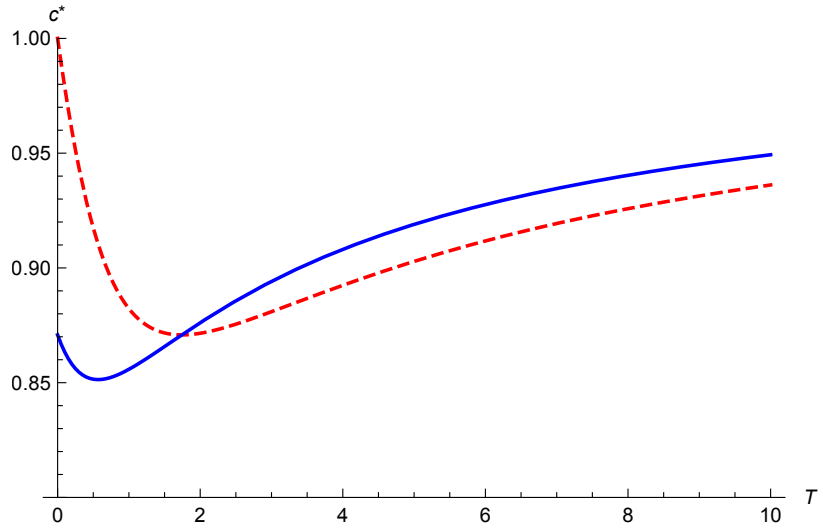
There exists a T^* such that equilibrium complexity is inefficiently high compared to the social planner's solution if and only if attention is abundant $T > T^*$.

The result that equilibrium complexity is too high if $\frac{T/N}{c} > 1$ is the equilibrium counterpart to the local deviation described in Proposition 4, which states that suppliers have an incentive to make goods too complex if they are well understood. A particularly interesting observation, illustrated in Figure 2, is that, for relatively high consumer attention budgets T, the equilibrium level of complexity (solid blue line) lies above that chosen by the planner (dashed red line). Therefore, complexity rises to inefficiently high levels precisely when information processing capacity grows. This result highlights a *complexity paradox*: Increased information processing capacity can be a source of excessive complexity in the economy because it induces individual suppliers to increase complexity above the efficient level. This finding echoes Ellison and Wolitzky (2012), who show that, in a search framework, reductions in consumer search costs are partially offset by increased obfuscation by firms.

Figure 2 and Proposition 5 together highlight the importance of distinguishing between absolute and relative complexity. For example, while a decrease in the attention budget T (or an increase in the number of goods N) leads to overly simple goods relative to the social planner's solution, it is not necessarily the case that goods become simpler in an absolute sense. Figure 2 illustrates that decreasing the attention budget T can lead to an increase in absolute complexity. The reason is that, when attention is severely limited (low T), no goods are well understood, so that it becomes optimal for suppliers to focus solely on the direct benefit of complexity.

Finally, note that Proposition 5 assumes that the benefit of understanding the good δ is not too large. This assumption ensures that the equilibrium is symmetric. When δ is large, an asymmetric equilibrium might arise: For high δ and small attention capacity T it can be optimal (both in equilibrium and in the social planner's solution) to choose complexities asymmetrically across otherwise ex-ante identical goods: One good is very simple ($c_i = 0$)

Figure 2: **Equilibrium and socially optimal complexity as a function of attention capacity** T



The figure shows equilibrium complexity (solid blue line) and the complexity chosen by a constrained planner (dashed red line) as a function of the attention budget T. Both equilibrium and constrained optimal complexity converge to the unconstrained optimal level of complexity of 1 as $T \to \infty$. Homogeneous goods with parameters $N = 2$, $\delta = 0.9$, $\alpha = 2$, $\theta = 0.5$.

and receives no attention, whereas the other good is complex ($c_j > 0$) and receives all of the consumer's attention. In this case, fundamentally similar goods can have very different levels of complexity.

4 Discussion of Assumptions

In this section, we provide a discussion of some of the assumptions underlying our model.

Market power. As noted in the introduction, some degree of market power is essential to our results. Specifically, suppliers must be able to extract a portion of the surplus created when consumers allocate attention to their goods. In the baseline model, we assume that each supplier is a monopolist in its own good. While literal monopoly is not required, perfect competition would restore efficiency. For example, under Bertrand competition, competing

suppliers would compete on complexity, driving it to the level preferred by the consumer, which coincides with the social planner's solution.

Endogenous surplus sharing. A key simplifying assumption in our analysis is that the supplier receives a fixed share θ_i of the value of a unit good. While this reduced-form specification is realistic in some applications (e.g., the instructor example in the introduction), it raises the question of whether prices can eliminate the complexity externality when goods are traded in markets.

In Appendix B, we show that the complexity inefficiency persists in a standard consumer-theory setting with endogenous prices and continuous quantity choice. The key insight is that the good's price is a one-dimensional instrument that cannot simultaneously address both quantity and complexity choice. The complexity externality therefore survives even when goods prices are determined in equilibrium. In this setting, eliminating the externality would require a separate price for attention.

More generally, in settings in which the supplier and consumer bargain over the surplus generated by the good, restoring efficiency requires that bargaining is over the *net* surplus generated by the good. This requires that the consumer can credibly threaten not to consume the good and to reallocate all attention to other goods. In settings in which some attention is sunk at the time of bargaining or when consumers cannot credibly promise not to consume a good (e.g., because an Inada condition makes this too costly), the complexity inefficiency remains.

Limited attention. Another assumption of our model is that limited attention takes the form of a hard constraint on the consumer's time budget. This is meant to capture that the amount of time that a consumer can spend on analyzing and consuming goods is limited. This assumption is uncontroversial for individuals who have to allocate their time to understand several goods. In the case of companies, one may argue that, by employing more people or purchasing additional information technology, the company can relax its attention constraint.

However, as long as increasing the attention budget is costly, the implications of such a setting would be similar to those under a hard attention constraint.[12]

5 Applications

In this section, we discuss two applications of our model: (i) regulation and (ii) communication. We discuss further applications with continuous quantity choice in Appendix B.

5.1 Financial Regulation

It is common for firms to spend a substantial amount of time to ensure they comply with regulation. Thus regulators compete for attention with other operations of the firms. In fact, in many cases, firms face multiple regulators, in which case regulators also compete for attention with each other. One prime example is the regulation of financial institutions, such as banks, insurance companies, and asset managers, where different market segments and activities fall under the remit of different supervisors (e.g., the Federal Reserve, CFTC, SEC, OCC). Even within a single regulator, different parts of the regulatory framework are usually drafted by separate committees and subcommittees.

The mapping from our model to financial regulation is as follows. The consumer is a financial institution that must allocate a limited budget of time that can be devoted to compliance, T, across multiple regulatory domains. The suppliers are regulators, each designing rules for their specific regulatory domain. The complexity c_i of a regulatory framework reflects its level of detail, the number of provisions, and the granularity of reporting requirements. More complex regulation can improve outcomes within a given domain by targeting specific risks more precisely and closing regulatory loopholes, capturing the direct benefit of complexity $f(c_i)$. At the same time, understanding and complying with more complex regulation requires more of

[12] If the cost of additional attention capacity is fully borne by the consumer but she only captures part of the social benefit, this introduces underinvestment in attention as an additional source of inefficiency not present in our model.

the institution's limited compliance resources. The depth of understanding $d_i = t_i/c_i$ corresponds to the quality of the institution's compliance with regulation i. Each regulator cares primarily about the surplus generated within its own regulatory domain, such as financial stability or consumer protection in a specific market segment, which corresponds to the supplier's objective $\theta_i \cdot v_i$. The regulated institution also benefits from operating in a well-regulated market, justifying a payoff proportional to v_i.[13] The attention externality arises because each regulator, when increasing the complexity of its own framework, does not internalize that doing so draws compliance resources away from other regulatory domains (and from the institution's own operations).

Our model identifies a specific channel through which regulatory complexity becomes excessive: each regulator has an incentive to increase the granularity and detail of its own rules to improve outcomes in its regulatory domain, without accounting for the resulting burden on the institution's ability to comply with other regulations or to conduct its core business. This mechanism operates even when each individual regulation is well-intentioned and regulatory complexity genuinely improves outcomes within a given domain.

Our model implies that in such a situation the complexity of financial regulation will generally not coincide with the social optimum. Given the high stakes and the potential for fines, banks need to understand regulation well in order to comply. According to Propositions 4 and 5, this is likely to result in financial market regulation that is excessively complex.[14]

These results provide a novel angle to the ongoing debate on regulatory complexity (see, e.g., Gai et al., 2019). Specifically, the prediction that equilibrium regulatory complexity is likely to be too high when regulators design regulations without internalizing that they draw

[13] In this interpretation, the weights assigned by the regulator and the financial institution to the value created in a given regulatory domain need not sum to one, as they reflect the importance each party assigns to market outcomes rather than a surplus split.

[14] Of course, the attention budget of a financial institution is not literally fixed. Rather, financial institutions react by expanding personnel to deal with excessively complex regulation. The inefficiency in this case is the opportunity cost generated when the employees required to deal with bank regulation could be engaged in more productive activities.

on a financial institution's limited attention budget is in line with recent evidence that has documented that the fragmented U.S. regulator landscape is indeed characterized by increasingly complex regulation. For example, Haldane and Madouros (2012) document increased complexity as measured by the word count of financial regulation. Herring (2018) documents that globally significant banks now have to meet an increasing number of regulatory capital requirements. Colliard and Georg (2025) develop a methodology to measure regulatory complexity analogous to measures of algorithmic complexity in computer science.

The mechanism that leads to excessive regulatory complexity in our model is different from (and likely complementary to) other channels that have been proposed in the literature. For example, in Hakenes and Schnabel (2014), sophisticated banks can capture the regulator by complex arguments that the regulator does not fully understand. In Asriyan, Foarta and Vanasco (2023), a regulatory proposal has to be approved or rejected by the median voter. When the proposed regulation is complex, the decision to approve is made mainly based on the median voter's prior. From the supplier's perspective, complex regulation is optimal if, for example, there is strong demand for new regulation.

Applying the same rationale to everyday life, businesses and households confront similar issues: Laws and regulations (tax law, traffic rules, parking regulations, etc.) all draw on our limited attention. Therefore, equilibrium complexity is likely inefficient. Indeed, Zwick (2021) documents that the complexity of the corporate tax code leads to suboptimal behavior and has significant costs for firms.

5.2 Communication

The results of our model apply in any setting in which suppliers of messages compete for the attention of a single party. A salient example is communication within a firm. The consumer is a CEO who must allocate a limited time budget T across issues arising in multiple divisions. Each division head is a supplier who chooses the complexity c_i of the report or briefing sent

to the CEO, and the depth of understanding $d_i = t_i/c_i$ reflects how well the CEO grasps the issues in division i. Each division head will care about the value to the CEO of the report sent by their particular division, ignoring that their report will affect how much time the CEO has left to deal with issues that arise in other divisions. If a more complex report induces the CEO to allocate more time to that division, division heads have an incentive to prepare reports that are overly complex. According to Propositions 4 and 5, this will especially be the case for issues that are well understood by the CEO. Analogously, if a simpler report induces the CEO to allocate more time, issues will be communicated in an oversimplified manner. Hence, our model predicts that simple issues that are easy to understand are blown up and communicated in an overly complex way, whereas difficult issues are communicated in an overly simple manner. This result also suggests that it can be efficient for CEOs to allocate a fixed amount of time to certain issues in order to counteract the attention externality.

Similar effects arise in other settings in which multiple actors communicate with a single attention-constrained individual or entity. Consider how politicians or lobby groups communicate their issues to citizens. When designing their messages, politicians and lobby groups have in mind mainly their own narrow issues and agendas, whereas citizens have to allocate their attention across the many separate issues that invariably affect them. Given the limited time citizens can devote to understanding these issues, intricate policy debates will, in general, not be well understood. Propositions 4 and 5 then imply that politicians, lobby groups, and the media will oversimplify these issues when communicating to the public. For example, despite the apparent complexities, the question of whether the UK should leave the EU was often oversimplified to a question about the UK's gross contribution to the EU budget. Correspondingly, smaller and relatively simple issues that are well understood by citizens are likely to be presented in an overly complicated fashion in order to attract attention.

Irrespective of whether the equilibrium distortion features too much or too little complexity, Proposition 2 implies that the chosen complexity of messages leads to an inefficient burden

on consumers' attention and, consequently, a welfare loss. Relative to the efficient allocation, CEOs will feel pressed for time and citizens overwhelmed by the issues they face.

6 Addressing the Tragedy of Complexity

How can the tragedy of complexity be addressed? The standard approach to common pool problems is to price the shared resource. However, directly pricing consumer attention is typically infeasible. In some instances, the price of the good itself can help overcome the inefficiency, as discussed in Section 4. This requires that all consumer attention is allocated after prices are determined. If some attention is sunk before prices are set, the inefficiency persists in the form of a hold-up problem, even when goods are priced. Moreover, in many salient cases, including regulation and communication, goods themselves are not priced at all.

Given that some degree of market power is essential to our results, increased competition between suppliers could help in principle: as discussed in Section 4, perfect competition would restore efficiency by eliminating suppliers' ability to extract surplus. When the perfect competition benchmark cannot be achieved, two alternative approaches are available in non-price settings.

First, efficiency can be improved by allowing consumers to choose the complexity of the goods they consume. In the symmetric case, when the fraction of value extracted by consumers θ is equal across goods, the consumer's objective function (3) and the social planner's objective (10) are identical up to a scaling factor. Therefore, consumers would choose the socially optimal level of complexity. However, even in this symmetric case, individual suppliers have an incentive to deviate from offering the optimal complexity. Following the logic of Proposition 1, if all other suppliers offer the efficient complexity level, an individual supplier benefits from deviating. This solution therefore requires coordinated action and binding regulation to prevent such deviations.

Second, based on Proposition 2, consumers can restore efficiency by committing ex ante to allocate a fixed amount of time to each good. For example, CEOs may commit to spending predetermined amounts of time with each division. Such commitment eliminates the reallocation of attention in response to complexity changes and, hence, the attention externality. With fixed time allocations, suppliers' first-order conditions coincide with the social planner's, restoring efficiency. However, this solution demands substantial sophistication: consumers must correctly determine and commit to the efficient time allocation for each good.

7 Conclusion

In many settings, competing suppliers draw on the limited attention that an economic agent allocates across multiple goods and activities. Examples include financial institutions facing regulatory bodies and CEOs interacting with corporate divisions.

This paper shows that, in such situations, equilibrium complexity is generally inefficient, even though complexity itself can be value-enhancing. The reason for this inefficiency is an attention externality. When choosing complexity, suppliers do not take into account that consumer attention is a common resource shared across goods. Suppliers distort complexity to increase attention paid to their goods, thereby diverting attention away from other goods or activities. Depending on the consumer's reaction to an increase in complexity—does the consumer devote more or less time when a good becomes more complex?— this leads to *too much or too little complexity* in equilibrium.

Our analysis shows that, under reasonably general conditions, consumer attention is hump-shaped in complexity. Hump-shaped attention allocation implies that firms overcomplicate goods that are well-understood in the social planner's solution. Conversely, firms oversimplify goods that, according to the planner, do not need to be well understood by consumers. Finally, equilibrium complexity is more likely to be excessive when attention is abundant. Therefore,

while advances in information processing capacity help consumers deal with existing complexity, they also make it more likely that equilibrium complexity is excessive.

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A Proofs

Proof of Lemma 1.

First note that raising the shadow cost of attention leads to less attention being paid to all goods. Rewriting the FOC for attention allocation (5) in terms of $\tilde{v}$ when it is binding yields

$$(1-\theta_i)\cdot\frac{\partial\tilde{v}_i\left(c_i,t_i(c_i,\lambda)\right)}{\partial t_i}=\lambda, \tag{A1}$$

taking total derivative with respect to λ:

$$\frac{\partial\tilde{t}_i\left(c_i,\lambda\right)}{\partial\lambda}=\frac{1}{1-\theta_i}\cdot\frac{1}{\frac{\partial^2\tilde{v}_i\left(c_i,\tilde{t}_i(c_i,\lambda)\right)}{\partial\tilde{t}_i^2}}<0, \tag{A2}$$

where we used $\tilde{v}_j(c_j,\tilde{t}_j)=v_j(c_j,\frac{t_j}{c_j})$ and $\frac{\partial^2 v_j(c_j,t_j)}{\partial {t_j}^2}=\frac{\partial^2 v_j\left(c_j,\frac{t_j}{c_j}\right)}{\partial\left(\frac{t_j}{c_j}\right)^2}\cdot\frac{1}{c_j^2}<0$, which holds by Assumption 1.

Equation (5) implicitly defines the attention allocated to good i, t_i, as $\tilde{t}_i\left(c_i,\lambda\right)$. Attention grabbing $\frac{\partial t_i(c_i,c_{-i})}{\partial c_i}$ can then be written as:

$$\frac{\partial t_i(c_i,c_{-i})}{\partial c_i}=\frac{d\tilde{t}_i(c_i,\lambda(c_i,c_{-i}))}{dc_i}=\frac{\partial\tilde{t}_i(c_i,\lambda)}{\partial c_i}+\frac{\partial\tilde{t}_i(c_i,\lambda)}{\partial\lambda}\cdot\frac{\partial\lambda(c_i,c_{-i})}{\partial c_i}, \tag{A3}$$

where the first term is the effect of c_i on t_i keeping λ fixed, while the second term captures the indirect effect through the shadow price of attention λ.

The equilibrium shadow price $\lambda(c_i,c_{-i})$ is implicitly defined by the binding attention constraint

$$T=\sum_j t_j=\sum_j\tilde{t}_j\left(c_j,\lambda\right). \tag{A4}$$

Without a specific functional form for v we cannot express $\lambda(c_i,c_{-i})$ explicitly. However, we can take a total derivative of (A4) with respect to c_i to get:

$$0=\frac{\partial\tilde{t}_i\left(c_i,\lambda\right)}{\partial c_i}+\sum_{j=1}^{N}\frac{\partial\tilde{t}_j\left(c_j,\lambda\right)}{\partial\lambda}\cdot\frac{\partial\lambda}{\partial c_i}, \tag{A5}$$

from which it follows that

$$\frac{\partial \lambda}{\partial c_i} = \frac{\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i}}{\sum_{j=1}^{N} -\frac{\partial \tilde{t}_j(c_j,\lambda)}{\partial \lambda}}. \tag{A6}$$

Plugging this into (A3) yields

$$\frac{\partial t_i(c_i,c_{-i})}{\partial c_i} = \frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i} + \frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial \lambda} \cdot \frac{\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i}}{\sum_{j=1}^{N} -\frac{\partial \tilde{t}_j(c_j,\lambda)}{\partial \lambda}} = \frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i} \cdot \left[1 - \frac{-\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial \lambda}}{\sum_{j=1}^{N} -\frac{\partial \tilde{t}_j(c_j,\lambda)}{\partial \lambda}}\right] \tag{A7}$$

The second term is positive as $\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial \lambda} < 0$ for all $i \in \{1, N\}$ (see (A2)). Thus $\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i}$ has the same sign as $\frac{\partial t_i(c_i,c_{-i})}{\partial c_i}$, proving the equivalence of (i) and (iii). By the same argument, it is obvious from (A6) that $\frac{\partial \lambda}{\partial c_i}$ also has the same sign as $\frac{\partial t_i(c_i,c_{-i})}{\partial c_i}$, proving the equivalence of (i) and (ii).

We now turn to the equivalence of (i) and (iii). We first rewrite the consumer's problem (3) in terms of $\tilde{v}$:

$$\max_{t_1,..t_N} \sum_{i=1}^{N} (1-\theta_i) \cdot \tilde{v}_i\left(c_i, t_i\right), \tag{A8}$$

which is maximized subject to the attention constraint (4). This yields the counterpart of Equation (5), which for interior solutions for t_i can be written as

$$\frac{1}{1-\theta_i} \cdot \lambda = \frac{\partial \tilde{v}_i\left(c_i, t_i\right)}{\partial t_i}. \tag{A9}$$

Differentiating this expression with respect to c_i (keeping all c_j, $j \neq i$, fixed), we obtain:

$$\frac{1}{1-\theta_i} \cdot \frac{\partial \lambda}{\partial c_i} = \frac{\partial^2 \tilde{v}_i\left(c_i, t_i\right)}{\partial c_i \partial t_i} + \frac{\partial^2 \tilde{v}_i\left(c_i, t_i\right)}{\partial t_i^2} \cdot \frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i} + \frac{\partial^2 \tilde{v}_i\left(c_i, t_i\right)}{\partial t_i^2} \cdot \frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial \lambda} \cdot \frac{\partial \lambda}{\partial c_i} \tag{A10}$$

where we take into account that t_i is a function of c_i and λ, $\tilde{t}_i(c_i,\lambda)$ and that λ is a function of all c_i.

From (A2) we know that $\frac{\partial^2 \tilde{v}_i(c_i,t_i)}{\partial t_i^2} \cdot \frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial \lambda} = 1$, so that:

$$\frac{\theta_i}{1-\theta_i} \cdot \frac{\partial \lambda}{\partial c_i} = \frac{\partial^2 \tilde{v}_i\left(c_i, t_i\right)}{\partial c_i \partial t_i} + \frac{1}{\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial \lambda}} \cdot \frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i}. \tag{A11}$$

Using (A6) to substitute $\frac{\partial \lambda}{\partial c_i}$, we then obtain

$$\frac{\theta_i}{1-\theta_i} \cdot \frac{\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i}}{\sum_{j=1}^{N} -\frac{\partial \tilde{t}_j(c_j,\lambda)}{\partial \lambda}} = \frac{\partial^2 \tilde{v}_i\left(c_i,t_i\right)}{\partial c_i \partial t_i} + \frac{1}{\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial \lambda}} \cdot \frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i}, \tag{A12}$$

which can be rearranged to yield

$$\frac{\frac{\theta_i}{1-\theta_i} \cdot \left(-\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial \lambda}\right) + \sum_{j=1}^{N} -\frac{\partial \tilde{t}_j(c_j,\lambda)}{\partial \lambda}}{\left(-\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial \lambda}\right) \cdot \sum_{j=1}^{N} -\frac{\partial \tilde{t}_j(c_j,\lambda)}{\partial \lambda}} \cdot \frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i} = \frac{\partial^2 \tilde{v}_i\left(c_i,t_i\right)}{\partial c_i \partial t_i}. \tag{A13}$$

From (A2) we know that $\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial \lambda} < 0$ for all $i \in \{1,..N\}$, which implies that $\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i}$ and $\frac{\partial^2 \tilde{v}_i(c_i,t_i)}{\partial c_i \partial t_i}$ have the same sign. □

Proof of Proposition 1.

The incentives of supplier i to change the level of complexity c_i of good i starting from the social planner's optimum $(c_1^*,...c_N^*)$ depend on the difference between the supplier's first-order condition (9) and that of the planner (12), both evaluated at the social planner's choice $(c_1^*,...c_N^*)$. We rewrite the difference between the right-hand side of the two first-order conditions as

$$\sum_{j \neq i} \frac{\lambda^*}{1-\theta_j} \cdot \frac{\partial t_j(c_1^*,...,c_N^*)}{\partial c_i} = \sum_{j \neq i} \frac{\lambda^*}{1-\theta_j} \cdot \frac{d\tilde{t}_j(c_j^*,\lambda(c_1^*,...,c_N^*))}{dc_i} = \frac{\partial \lambda(c_1^*,...,c_N^*)}{\partial c_i} \cdot \sum_{j \neq i} \frac{\lambda^*}{1-\theta_j} \cdot \frac{\partial \tilde{t}_j(c_j^*,\lambda^*)}{\partial \lambda}. \tag{A14}$$

The first step in (A14) uses the fact that we can rewrite the consumer's attention choice $t_j(c_1^*,..c_N^*)$ as a function of the complexity of good j and the shadow cost of attention λ, $\tilde{t}_j(c_j^*,\lambda(c_1^*,..c_N^*))$. The second step applies the chain rule, $\frac{d\tilde{t}_j(c_j^*,\lambda(c_1^*,..c_N^*))}{dc_i} = \frac{\partial \tilde{t}_j(c_j^*,\lambda^*)}{\partial \lambda} \cdot \frac{\partial \lambda(c_1^*,..c_N^*)}{\partial c_i}$ and moves the first (constant) term outside of the summation sign. λ^* denotes the shadow cost of attention at the social planner's solution, $\lambda(c_1^*,..c_N^*)$.

Therefore, from (A14) we see that the externality is negative if $\frac{\partial \lambda(c_1^*,...,c_N^*)}{\partial c_i} > 0$, meaning that the social planner's optimum must entail a lower level of c_i, which in turn increases the left hand side of (12) (due to the decreasing benefits of complexity we have assumed). We show that $\frac{\partial \lambda(c_1^*,...,c_N^*)}{\partial c_i}$ has the same sign as $\frac{\partial t_i(c_1^*,..c_N^*)}{\partial c_i}$ in the proof of Lemma 1. □

Proof of Proposition 2.

As shown in Proposition 1, the direction of the complexity distortion is determined by the sign of $\frac{\partial t_i(c_i,c_{-i})}{\partial c_i}$. Lemma 1 shows that this distortion has the same sign as $\frac{\partial \lambda}{\partial c_i}$. Therefore, suppliers have an incentive to distort complexity in the direction that raises the shadow price of attention λ. Thus, starting from the social planner's solution and allowing suppliers to follow a best-response strategy, in every step of the iteration the shadow cost of attention weakly increases (strictly if $\frac{\partial t_i(c_i,c_{-i})}{\partial c_i} > 0$). Thus $\lambda^e > \lambda^*$ whenever equilibrium complexity and the social planner's solution do not coincide. (They coincide if only if $\frac{\partial t_i(c_i,c_{-i})}{\partial c_i} = 0$ for all suppliers at the social planner's solution.) □

Proof of Lemma 2.

Using the chain rule

$$\frac{\partial^2 \tilde{v}(c_i,t_i)}{\partial c_i \partial t_i} = \frac{\partial^2 v(c_i,t_i/c_i)}{\partial c_i \partial t_i} = \frac{1}{c_i} \cdot \frac{\partial^2 v(c_i,t_i/c_i)}{\partial c_i \partial d_i} - \frac{1}{c_i^2} \cdot \frac{\partial v(c_i,t_i/c_i)}{\partial d_i} - \frac{t_i}{c_i^3} \cdot \frac{\partial^2 v(c_i,t_i/c_i)}{\partial d_i^2}. \tag{A15}$$

From Lemma 1 we know that $\frac{\partial t_i(c_i,c_{-i})}{\partial c_i}$ and $\frac{\partial^2 \tilde{v}(c_i,t_i)}{\partial c_i \partial t_i}$ have the same sign. Thus $\frac{\partial t_i}{\partial c_i} > 0$ if and only if

$$-\frac{\frac{\partial v_i}{\partial d_i}}{\frac{\partial^2 v_i}{\partial d_i^2} \cdot d_i} \cdot \left(1 - \frac{\frac{\partial^2 v_i}{\partial c_i \partial d_i} \cdot c_i}{\frac{\partial v_i}{\partial d_i}}\right) < 1, \tag{A16}$$

and $\frac{\partial t_i}{\partial c_i} < 0$ if and only if the opposite is true. Plugging in v_i for the two functional forms yields the result stated in the proposition. □

Proof of Lemma 3. By definition:

$$\frac{dd_i}{dc_i} = \frac{d}{dc_i}\left[\frac{t_i(c_i)}{c_i}\right] = \frac{\partial t_i}{\partial c_i} \cdot \frac{1}{c_i} - \frac{t_i}{c_i^2}. \tag{A17}$$

Thus if $\frac{\partial t_i}{\partial c_i} < 0$ then $\frac{\partial d_i}{\partial c_i} < 0$ trivially, so we only need to focus on the case of $\frac{\partial t_i}{\partial c_i} > 0$.

Taking the total derivative $\frac{d}{dc_i}$ (allowing d_i to adjust endogenously) of the first-order condition of attention allocation (5) for $t_i > 0$ (and thus $d_i > 0$), we get

$$(1-\theta_i) \cdot \left[\frac{\partial^2 v_i\left(c_i, \frac{t_i}{c_i}\right)}{\partial c_i \partial d_i} + \frac{\partial^2 v_i\left(c_i, \frac{t_i}{c_i}\right)}{\partial d_i^2} \cdot \frac{\partial d_i}{\partial c_i}\right] = \frac{\partial \lambda}{\partial c_i} \cdot c_i + \lambda. \tag{A18}$$

Rearranging we get:

$$\frac{\partial d_i}{\partial c_i} = \frac{\frac{1}{1-\theta_i} \cdot \left[\frac{\partial \lambda}{\partial c_i} \cdot c_i + \lambda\right] - \frac{\partial^2 v_i(c_i,d_i)}{\partial c_i \partial d_i}}{\frac{\partial^2 v_i(c_i,d_i)}{\partial d_i^2}}. \tag{A19}$$

Plugging $v_i(c,d) = f_i(c) + g_i(d)$ from Assumption 1 into (A19) we arrive at:

$$\frac{\partial d_i}{\partial c_i} = \frac{\frac{1}{1-\theta_i} \cdot \left[\frac{\partial \lambda}{\partial c_i} \cdot c_i + \lambda\right]}{g_i''(d_i)}. \tag{A20}$$

We know that $\lambda > 0$ always and $g_i''(d_i) < 0$ by definition. From Lemma 1 we know that $\frac{\partial \lambda}{\partial c_i} > 0$ if $\frac{\partial t_i}{\partial c_i} > 0$, thus it follows that $\frac{\partial d_i}{\partial c_i} < 0$ if $\frac{\partial t_i}{\partial c_i} > 0$, which was the case we had to prove.

□

Proof of Proposition Proposition 3.

We first prove $\lim_{c_i \to 0^+} t_i = 0$ by contradiction: Assume the opposite, i.e., that there exists a strictly decreasing infinite series $\lim_{n\to\infty} c_{i,n} = 0$ such that $\liminf_{n\to\infty} t_i(c_{i,n}) = \underline{t} > 0$. Define a new function $\tilde{t}_i = \frac{t_i}{2}$. With this new function we have $\liminf_{n\to\infty} \frac{\tilde{t}_i(c_{i,n})}{c_{i,n}} = \infty$ as well, thus as $c_i \to 0^+$ there is no loss in good i's value but at least $\frac{\underline{t}}{2}$ can be reallocated to other goods, yielding extra value of at least $\frac{\underline{t}}{2} \cdot \underline{\lambda}$ where $\underline{\lambda}$ is the shadow cost of attention if no attention was paid to good i. Thus the original $t_i(c_i)$ function was not an optimal choice of the consumer.

We now show that $\lim_{c_i \to \infty} t_i = 0$: By Assumption 1, we know that $v_i(c_i, d_i)$ is bounded from above and below thus $\lim_{d_i \to 0^+} d_i \cdot v_i(c_i, d_i) = 0$. Using l'Hôpital's rule:

$$\lim_{d_i \to 0^+} v_i(c_i, d_i) = \lim_{d_i \to 0^+} \frac{d_i \cdot v_i(c_i, d_i)}{d_i} = \lim_{d_i \to 0^+} \frac{v(c_i, d_i) + d_i \cdot \frac{\partial v_i(c_i,d_i)}{\partial d_i}}{1} = \lim_{d_i \to 0^+} v_i(c_i, d_i) + \lim_{d_i \to 0^+} \frac{\partial v_i(c_i, d_i)}{\partial d_i} \cdot d_i \tag{A21}$$

implying $\lim_{d_i \to 0^+} \frac{\partial v_i(c_i,d_i)}{\partial d_i} \cdot d_i = 0$ for any given c_i. Now we show that this convergence is uniform over any c_i: For additive v_i, we have $\lim_{d_i \to 0^+} \frac{\partial v_i(c_i,d_i)}{\partial d_i} \cdot d_i = \lim_{d_i \to 0^+} g'(d_i) \cdot d_i$ which does not even depend on c_i so is trivially uniformly convergent. In case of multiplicative v_i, we have $\lim_{d_i \to 0^+} \frac{\partial v_i(c_i,d_i)}{\partial d_i} \cdot d_i = \lim_{d_i \to 0^+} f(c_i) \cdot g'(d_i) \cdot d_i$ which again is also uniformly convergent as $f(c_i)$ is bounded from both below and above. Note that $\lim_{c_i \to \infty} \frac{\partial v_i(c_i,d_i)}{\partial d_i} \cdot d_i$ also exists for all d_i: For additive v_i, we have $\lim_{c_i \to \infty} \frac{\partial v_i(c_i,d_i)}{\partial d_i} \cdot d_i = g'(d_i) \cdot d_i$, while for multiplicative v_i we have $\lim_{c_i \to \infty} \frac{\partial v_i(c_i,d_i)}{\partial d_i} \cdot d_i = g'(d_i) \cdot d_i \cdot \lim_{c_i \to \infty} f(c_i)$, where the latter limit exists since $f(c_i)$ is increasing and bounded from above.

Thus we can use the Moore-Osgood theorem, see Taylor (2012):

$$\lim_{\substack{c_i\to\infty\\ d_i\to 0^+}} d_i \cdot \frac{\partial v\left(c_i, d_i\right)}{\partial d_i} = \lim_{c_i\to\infty}\lim_{d_i\to 0^+} \frac{\partial v_i\left(c_i, d_i\right)}{\partial d_i} \cdot d_i = 0 \tag{A22}$$

Since $t_i \in [0, T]$ by definition, we have $\lim_{c_i\to\infty} d_i(c_i) = \lim_{c_i\to\infty} \frac{t_i}{c_i} = 0$ and thus:

$$\lim_{c_i\to\infty} d_i \cdot \frac{\partial v\left(c_i, d_i(c_i)\right)}{\partial d_i} = \lim_{\substack{c_i\to\infty\\ d_i\to 0^+}} d_i \cdot \frac{\partial v\left(c_i, d_i\right)}{\partial d_i} = 0 \tag{A23}$$

Using the FOC of attention allocation (5) for any given c_i it follows that:

$$\lim_{c_i\to\infty} d_i(c_i) \cdot \frac{\partial v\left(c_i, d_i(c_i)\right)}{\partial d_i} = \lim_{c_i\to\infty} \frac{t_i(c_i)}{c_i} \cdot \frac{\partial v\left(c_i, d_i(c_i)\right)}{\partial d_i} = \frac{1}{1-\theta_i} \cdot \lim_{c_i\to\infty} t_i(c_i) \cdot \lambda = 0. \tag{A24}$$

Since λ is bounded from below by $\underline{\lambda} > 0$ that would prevail if no attention was allocated to good i, we conclude $\lim_{c_i\to\infty} t_i = 0$.

We now show that $\lim_{c_i\to 0^+} \frac{\partial t_i}{\partial c_i} > 0$: Note that from Lemma 3 d_i is strictly decreasing in c_i, thus $\lim_{c_i\to 0^+} d_i > 0$ unless $t_i(c_i) = 0$ for all c_i. Using $\lim_{c_i\to 0^+} t_i = 0$, by l'Hôpital's rule:

$$\lim_{c_i\to 0^+} d_i = \lim_{c_i\to 0^+} \frac{t_i}{c_i} = \lim_{c_i\to 0^+} \frac{\frac{\partial t_i}{\partial c_i}}{1} = \lim_{c_i\to 0^+} \frac{\partial t_i}{\partial c_i}. \tag{A25}$$

Since $\lim_{c_i\to 0^+} d_i > 0$, it follows that $\lim_{c_i\to 0^+} \partial t_i/\partial c_i > 0$.

We now show $t_i(c_i, c_{-i}) = 0$ for $c_i > \bar{\bar{c}}_i$: Note that all statements for $c_i > \bar{c}_i$ assume that the first-order condition (5) defining t_i is binding and thus $t_i > 0$. Now assume that (5) is not binding for certain $c_i > 0$ and denote the lim inf of such c_i's by $\bar{\bar{c}}_i$. Since v_i is continuously differentiable it follows that $t_i(c_i)$ defined with equality in (5) is continuous. Since $\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i} < 0$ for all $c_i > \bar{c}_i$, it follows that $\bar{\bar{c}}_i > \bar{c}_i$ and that (5) cannot be binding for any $c_i > \bar{\bar{c}}_i$ either. Thus $t_i(c_i) = 0$ for $c_i \geq \bar{\bar{c}}_i$ and thus $\frac{\partial \tilde{t}_i(c_i,\lambda)}{\partial c_i} = 0$ for $c_i > \bar{\bar{c}}_i$. From (5) and that $f_i(c_i)$ is increasing and bounded from above it follows that $\bar{\bar{c}}_i < \infty$ if and only if $\lim_{d_i\to 0^+} g_i'(d_i) < \infty$.

We now show there exists a $c_i > 0$ such that $\frac{\partial t_i}{\partial c_i} < 0$: First note that since $t_i \geq 0$ and $\lim_{c_i\to 0^+} \frac{\partial t_i}{\partial c_i} > 0$, there exists a $\tilde{c}_i > 0$ such that $t_i(\tilde{c}_i, c_{-i}) > 0$. Second note that $t_i(c_i, c_{-i})$ is defined by (5): Since v_i is continuously differentiable in c_i, it must be the case that $t_i(c_i, c_{-i})$ is continuous in c_i. Now assume

by contradiction that $\frac{\partial t_i}{\partial c_i} \geq 0$ always, in that case $\liminf_{c_i \to \infty} t_i(c_i, c_{-i}) \geq t_i(\tilde{c}_i, c_{-i}) > 0$ which directly contradicts the above result that $\lim_{c_i \to \infty} t_i = 0$.

Note that the proof is quite general until this point as we did not rely on any assumptions regarding the additivity of v_i or monotonicity of σ_i (more specifically on parts (i) and (vi) of Assumption 1).

We show that $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i} > 0$ for $c_i < \bar{c}_i$ and $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i} < 0$ for $c_i \in (\bar{c}_i, \bar{\bar{c}}_i)$: By Assumption 1, $\sigma_i(d_i)$ is strictly decreasing in d_i and by Lemma 3 d_i is strictly decreasing in c_i, thus we have that $\sigma_i(d_i(c_i))$ is strictly increasing in c_i. Since we have shown that $\lim_{c_i \to 0^+} \frac{\partial t_i}{\partial c_i} > 0$, Lemma 2 implies that $\lim_{c_i \to 0^+} \sigma_i(d_i(c_i)) < 1$. Previously we have shown that there is a $c_i \in (0, \infty)$ at which $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i} < 0$ thus by Lemma 2, at that specific c_i, we have $\sigma_i(d_i(c_i)) > 1$. Since $\sigma_i(d_i(c_i))$ is strictly increasing in c_i, there is a unique cutoff $\bar{c}_i \in (0, \infty)$ defined by $\sigma_i(d_i(\bar{c}_i)) = 1$. From Lemma 2 it then follows that $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i} > 0$ for $c_i < \bar{c}_i$ and $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i} < 0$ for $c_i \in (\bar{c}_i, \bar{\bar{c}}_i)$. Note that we also used the continuity of $\sigma_i(d_i(c_i))$ in c_i for the above proof which follows from g_i being continuously differentiable for $c_i > 0$ and also $d_i(c_i)$ being continuous.

□

Proof of Proposition 5.

First, we derive the equilibrium FOC's in a symmetric equilibrium. For expositional purposes we allow for all v_i's to be different. In case of an interior solution, the consumer's first-order condition for a given shadow price of attention λ yields:

$$\tilde{t}_i(c_i, \lambda) = \sqrt{\frac{c_i \cdot \delta_i \cdot (1 - \theta_i)}{\lambda}} - c_i. \tag{A26}$$

Plugging this expression into the (binding) attention constraint (4), we can express λ as:

$$\lambda = \frac{\left(\sum_{k=1}^{N} \sqrt{c_k \delta_k (1 - \theta_k)}\right)^2}{\left(\sum_{j=1}^{N} c_j + T\right)^2}. \tag{A27}$$

Substituting this expression back into Equation (A26) we obtain

$$t_i(c_1, ..c_N) = \frac{\sqrt{c_i \delta_i (1 - \theta_i)}}{\sum_{k=1}^{N} \sqrt{c_k \delta_k (1 - \theta_k)}} \cdot \left(\sum_{j=1}^{N} c_j + T\right) - c_i. \tag{A28}$$

Partially differentiating with respect to c_i yields

$$\frac{\partial t_i(c_1,..c_N)}{\partial c_i} = \frac{\sum_{j\neq i}\sqrt{c_j\delta_j\left(1-\theta_j\right)}}{\left(\sum_{k=1}^{N}\sqrt{c_k\delta_k\left(1-\theta_k\right)}\right)^2}\cdot\frac{1}{2}\cdot\frac{1}{\sqrt{c_i}}\cdot\sqrt{\delta_i\left(1-\theta_i\right)}\cdot\left(\sum_{j=1}^{N}c_j+T\right)+\frac{\sqrt{c_i\delta_i\left(1-\theta_i\right)}}{\sum_{k=1}^{N}\sqrt{c_k\delta_k\left(1-\theta_k\right)}}-1. \tag{A29}$$

Imposing symmetry ($c_i = c,\ \forall i$), this implies that

$$\frac{\partial t}{\partial c_i} = \frac{N-1}{2\cdot N}\cdot\left(\frac{T}{N\cdot c}-1\right) \tag{A30}$$

and

$$\lambda = \frac{\delta\cdot(1-\theta)}{c\cdot\left(1+\frac{T}{N\cdot c}\right)^2}. \tag{A31}$$

Plugging these into Equations (9) and (13), using the fact that under the assumed functional form $\frac{\partial v_i\left(c_i,\frac{t_i}{c_i}\right)}{\partial c_i} = f'(c)$, and observing that in symmetric equilibrium all goods get the same amount of attention $t = \frac{T}{N}$, we then arrive at the following equations.

In a symmetric equilibrium the equilibrium first-order condition is

$$\alpha - 2\cdot c = \left[\frac{T}{N\cdot c}-\frac{N-1}{2\cdot N}\cdot\left(\frac{T}{N\cdot c}-1\right)\right]\cdot\frac{\delta}{c\cdot\left(1+\frac{T}{N\cdot c}\right)^2}, \tag{A32}$$

whereas the social planner's first-order condition is

$$\alpha - 2\cdot c = \frac{T}{N\cdot c}\cdot\frac{\delta}{c\cdot\left(1+\frac{T}{N\cdot c}\right)^2}. \tag{A33}$$

Comparing (A32) with (A33) shows that equilibrium complexity c is higher than the social planner's choice of complexity if and only if $c < \frac{T}{N}$.

Second, we show that there exists a $\delta^* > 0$, such that for $\delta < \delta^*$ both the social planner's solution and the equilibrium outcome are symmetric.

We first calculate a strictly positive lower bound for δ^* that is sufficient to ensure symmetric complexity choices, both under the social planner's solution and in equilibrium. First note that the maximum value v that can be derived from a good is when the complexity is chosen to be the first best (unconstrained) optimum $c_i = \frac{\alpha_i}{2}$ and depth of understanding is chosen to be infinite $\frac{t_i}{c_i} = \infty$.

Therefore, v_i can be bounded from above,

$$v_i\left(c_i, \frac{t_i}{c_i}\right) < v_i\left(\frac{\alpha_i}{2}, \infty\right) = \left(\delta_i + \frac{\alpha_i^2}{4}\right). \tag{A34}$$

With N ex-ante symmetric goods, if all goods have the same amount of complexity and therefore receive the same amount of attention from the consumer, v_i can be bounded from below because if $c = \frac{\alpha}{2}$ is not optimal, then v must be higher under the optimal complexity choice:

$$v_i\left(c_i, \frac{t_i}{c_i}\right) > v\left(\frac{\alpha}{2}, \frac{T/N}{\alpha/2}\right) = \frac{\alpha^3 N + 2T\left(\alpha^2 + 4\delta\right)}{4\alpha N + 8T} \tag{A35}$$

To ensure that the social planner's choice of complexity is symmetric, it has to be the case that the social planner's first-order condition holds with equality for all goods. In particular, the planner must not have an incentive to set the complexity of one of the goods to zero. This is satisfied as long as

$$N \cdot v\left(c_s^*, \frac{t_{s,s}^*}{c_s^*}\right) > (N-1) \cdot v\left(c_a^*, \frac{t_{s,a}^*}{c_a^*}\right) + 1 \cdot v\left(0, \infty\right), \tag{A36}$$

where c_s^* is the social planner's optimum in the symmetric case and c_a^* in the asymmetric one (in which one of the goods has zero complexity but other goods are symmetric in complexity). From Equations (A34) and (A35), a sufficient condition for a symmetric solution is that

$$N \cdot \frac{\alpha^3 N + 2T\left(\alpha^2 + 4\delta\right)}{4\alpha N + 8T} > (N-1) \cdot \left(\delta + \frac{\alpha^2}{4}\right) + \delta, \tag{A37}$$

which simplifies to

$$\delta < \frac{\alpha(\alpha N + 2T)}{4N^2}. \tag{A38}$$

Note that this condition holds for small enough δ and is more likely to be violated when the attention capacity T is small. The condition holds for all $T > 0$ if $\delta < \frac{\alpha^2}{4 \cdot N}$. This is the sufficient (but clearly not necessary) condition for the social planner's solution to be symmetric.

Now we show that for low enough δ there exists a symmetric equilibrium in which all suppliers choose the complexity c_s^e. First note that from Equation (A32) it follows that if $\delta \to 0^+$ then $c_s^e \to \frac{\alpha}{2}$ from below. Thus for δ close enough to 0 it must hold that $\frac{\alpha}{4} < c_s^e < \frac{\alpha}{2}$. Because $v(.,.)$ is increasing in both arguments for $c < \frac{\alpha}{2}$, we can bound the supplier's payoff in a symmetric equilibrium from below.

Specifically,

$$v\left(c_s^e, \frac{t_s^e}{c_s^e}\right) > v\left(\frac{\alpha}{4}, \frac{T/N}{\alpha/2}\right) = \frac{3}{16} \cdot \alpha^2 + \delta \cdot \frac{\frac{2 \cdot T}{N \cdot \alpha}}{1 + \frac{2 \cdot T}{N \cdot \alpha}}. \tag{A39}$$

It remains to be shown that the supplier does not want to deviate from the symmetric equilibrium to producing a good with zero complexity (and thus infinite depth of understanding by the consumer). This requires that

$$v\left(c_s^e, \frac{t_s^e}{c_s^e}\right) > v(0, \infty) = \delta. \tag{A40}$$

Using Equation (A39) it suffices to show that

$$\frac{3}{16} \cdot \alpha^2 + \delta \cdot \frac{\frac{2 \cdot T}{N \cdot \alpha}}{1 + \frac{2 \cdot T}{N \cdot \alpha}} > \delta, \tag{A41}$$

which holds for any $T > 0$ if $\delta < \frac{3}{16} \cdot \alpha^2$. Thus we have shown that for small enough δ, there exists a symmetric equilibrium.

Third, we show that, starting from a set of parameters for which symmetric complexity choices coincide under the social planner's solution and in equilibrium, the comparative statics stated in the proposition hold. The separating hyperplane for which the optimal and equilibrium levels of complexity are the same happens at critical attention level T at which all goods have complexity $c = \frac{T}{N}$ (see Equation (17)). Plugging this into Equation (A33) yields a quadratic equation for T, for which the unique solution that corresponds to a maximum of the social welfare function is (this can be checked by signing the second order condition):

$$T^{crit} = \frac{N}{4}\left(\alpha + \sqrt{\alpha^2 - 2\delta}\right). \tag{A42}$$

Note that this critical T only exists if $\delta \leq \frac{\alpha^2}{2}$. Equation (A42) defines a separating hyperplane in the parameter space. By continuity of the equilibrium complexity in the underlying parameters, all we have to check is whether there is too much complexity on one side of the hyperplane, arbitrarily close to the hyperplane itself.

To prove that complexity is inefficiently large if attention is abundant, we take the total derivative of the two first-order conditions (A32) and (A33) with respect to T. Substituting $c = \frac{T}{N}$ (at $c^e = c^*$)

and solving for $\frac{dc^e}{dT}$ and $\frac{dc^*}{dT}$ yields:

$$\frac{dc^e}{dT} = \frac{\delta - \delta N}{\delta N(N+1) - 16T^2}, \tag{A43}$$

$$\frac{dc^*}{dT} = 0. \tag{A44}$$

We note that $\frac{dc^e}{dT} > \frac{dc^*}{dT}$ holds at $T = T^{crit}$ if $\delta < \frac{\alpha^2}{2}$ (i.e., if T^{crit} exists). □

B Microfoundation with priced good

In this section, we provide a microfoundation with a price for the simple model presented in Section 2. Recall that in our baseline model we assumed that the surplus generated by the good is split between the supplier and the consumer according to exogenous weights θ and $1-\theta$ and there is no price. The sharing of surplus is determined by an equilibrium price and the quantity consumed is a continuous choice. We show that the resulting first-order conditions with respect to complexity are exactly the same as those in the baseline model.

Payoffs. Suppose the consumer purchases q_i units of a divisible good i with associated complexity c_i at a unit price p_i. To put bounds on quantities, assume the unit cost for the supplier of producing good i is $k_i > 0$. The consumer devotes t_i units of time to understand the good. Assume that the consumer's valuation of good i takes the following functional form:

$$V_i\left(q_i, c_i, \frac{t_i}{c_i}\right) = u_i(q_i) \cdot v_i\left(c_i, \frac{t_i}{c_i}\right). \tag{B1}$$

The multiplicative structure of V_i ensures that there is no difference between the marginal and the "marginal marginal" value of complexity. This assumption ensures that no distortion à la Spence (1975) arises. We assume that the value of consuming each good i satisfies the Inada condition $\lim_{q_i \to 0^+} \frac{\partial}{\partial q_i} V_i\left(q_i, c_i, \frac{c_i}{t_i}\right) = \infty$. This simplifies the analysis by ensuring that all goods are consumed in equilibrium. We make the standard assumptions $u_i'(q_i) > 0$ and $u_i''(q_i) < 0$. Given the above multiplicative structure we also assume $u_i \geq 0$. Assumption 1 holds for v_i as before.

The above utility function nests a Dixit-Stiglitz specification as a special case. Setting $u_i(q_i) = q_i^\rho$ for all i with $\rho \in (0,1)$, the consumer's objective function becomes $\sum_i v_i\left(c_i, \frac{t_i}{c_i}\right) \cdot q_i^\rho - p_i \cdot q_i$.[15] This is equivalent to maximizing the Dixit-Stiglitz aggregator $\left(\sum_i v_i\left(c_i, \frac{t_i}{c_i}\right) \cdot q_i^\rho\right)^{1/\rho}$ subject to the budget constraint, where v_i serves as the preference weight for variety i. Crucially, these preference weights are endogenous: they depend on both the complexity chosen by suppliers and the attention allocated by consumers.

Timing. The timing of the game is the following: First the suppliers set complexities c_i and prices p_i of their goods, then the consumer chooses the quantity q_i to be consumed of good i along with the amount of attention t_i it devotes to good i. In the following, we solve for the equilibrium backwards.

Consumer. For each good i, the consumer chooses the quantity demanded q_i and attention t_i paid to the good by solving the following maximization problem:

$$\max_{q_1,..q_N,t_1..t_N} \sum_{i=1}^{N} u_i(q_i) \cdot v_i\left(c_i, \frac{t_i}{c_i}\right) - p_i \cdot q_i, \tag{B2}$$

subject to the attention constraint (4).

The first-order condition with respect to attention t_i paid to good i is given by

$$\lambda \geq u_i(q_i) \cdot \frac{\partial v_i\left(c_i, \frac{t_i}{c_i}\right)}{\partial d_i} \cdot \frac{1}{c_i}, \tag{B3}$$

which holds with equality whenever $t_i > 0$. The first-order condition with respect to the quantity q_i demanded of good i is given by

$$p_i = \frac{\partial u_i\left(q_i\right)}{\partial q_i} \cdot v_i\left(c_i, \frac{t_i}{c_i}\right). \tag{B4}$$

The first-order condition with respect to quantity demanded holds with equality given the Inada condition that ensures all goods will be consumed in positive quantity ($q_i > 0,\ \forall i$).

Supplier. The supplier of good i chooses the good's complexity c_i and its price p_i to maximize profits. When choosing complexity and price, the supplier takes into account its effect on quantity consumed, both directly through $q_i(c_i, p_i)$, and also indirectly through its effect on the amount of

[15] If s is the elasticity of substitution, then $\rho = \frac{s-1}{s}$.

attention paid to the good $t_i(c_i, p_i)$. Hence, the supplier's problem is given by

$$\max_{c_i, p_i} p_i \cdot q_i(c_i, p_i) - k_i \cdot q_i(c_i, p_i). \tag{B5}$$

The supplier's first-order condition with respect to the price p_i is given by

$$(p_i - k_i) \cdot \frac{\partial q_i}{\partial p_i} + q_i = 0. \tag{B6}$$

To have non-negative profits in equilibrium we need $p_i - k_i \geq 0$. However, $p_i - k_i = 0$ would imply $q_i = 0$ and thus we must have $p_i - k_i > 0$ in equilibrium. Therefore, suppliers make strictly positive profits because they are monopolists in their own good.

Now we turn to the complexity choice. For simplicity, we only look for equilibria with interior solutions in c_i. Recall that q_i will be interior in any case because of the Inada condition. The first-order condition with respect to complexity c_i is then given by

$$(p_i - k_i) \cdot \frac{\partial q_i}{\partial c_i} = 0. \tag{B7}$$

Since $p_i - k_i > 0$ the above implies $\frac{\partial q_i}{\partial c_i} = 0$.

The supplier knows that its choice of complexity c_i and price p_i affect the consumer's quantity $q_i(c_i, p_i)$ and time choice $t_i(c_i, p_i)$, which are pinned down by the equilibrium conditions (B3) and (B4). First we take the partial derivative of (B4) with respect to c_i, while keeping the price p_i fixed. We arrive at

$$0 = \frac{\partial^2 u_i\left(q_i\right)}{\partial {q_i}^2} \cdot \frac{\partial q_i}{\partial c_i} \cdot v_i\left(c_i, d_i\right) + \frac{\partial u_i\left(q_i\right)}{\partial q_i} \cdot \left[\frac{\partial v_i\left(c_i, d_i\right)}{\partial c_i} - \frac{\partial v_i\left(c_i, d_i\right)}{\partial d_i} \cdot \frac{1}{c_i}\left(\frac{t_i}{c_i} - \frac{\partial t_i}{\partial c_i}\right)\right], \tag{B8}$$

where we took into account that the consumer's equilibrium choice of q_i and t_i depends on the supplier's choice of c_i. Using $\frac{\partial q_i}{\partial c_i} = 0$ and given that both q_i and $\frac{\partial u_i(q_i)}{\partial q_i}$ are strictly positive, we get

$$\frac{\partial v_i\left(c_i, d_i\right)}{\partial c_i} - \frac{\partial v_i\left(c_i, d_i\right)}{\partial d_i} \cdot \frac{1}{c_i}\left(\frac{t_i}{c_i} - \frac{\partial t_i}{\partial c_i}\right) = 0. \tag{B9}$$

This condition is exactly the same as (8) for the baseline model in the paper.

Planner. We now derive the corresponding optimality condition for the planner's solution. To simplify the exposition, we allow the planner to directly choose quantities consumed q_i instead of choosing the price p_i. For each good i, the planner chooses complexity c_i and quantity q_i, taking into account their effect on attention $t_j(c_1, ..c_N, q_1, ..q_N)$ paid to all goods $j \in \{1, \ldots, N\}$ to maximize total surplus

$$\max_{c_1,..c_N,q_1,..q_N} \sum_{i=1}^{N} u_i(q_i) \cdot v_i \left(c_i, \frac{t_i(c_1, ..c_N, q_1, ..q_N)}{c_i} \right) - k_i \cdot q_i. \tag{B10}$$

The first-order condition with respect to quantity q_i is given by

$$\frac{\partial u_i}{\partial q_i} v_i - k_i + \sum_j u_j \frac{\partial v_j}{\partial d_j} \frac{1}{c_j} \frac{\partial t_j}{\partial q_i} = 0, \tag{B11}$$

where the summation term is equal to zero because $u_j \frac{\partial v_j}{\partial d_j} \frac{1}{c_j} = \lambda$ (from (B3)) and total attention is fixed thus attention can only be reallocated among goods $\sum_j \frac{\partial t_j}{\partial q_i} = 0$. Thus the first-order condition with respect to quantity simplifies to

$$\frac{\partial u_i}{\partial q_i} v_i = k_i. \tag{B12}$$

The first-order condition with respect to complexity c_i is given by

$$u_i(q_i) \frac{\partial v_i}{\partial c_i} - u_i(q_i) \frac{\partial v_i}{\partial d_i} \frac{1}{c_i} \frac{t_i}{c_i} + \sum_{j=1}^{N} u_j(q_j) \frac{\partial v_j}{\partial d_j} \frac{1}{c_j} \frac{\partial t_j}{\partial c_i} = 0. \tag{B13}$$

Given that (B3) implies $u_j \frac{\partial v_j}{\partial d_j} \frac{1}{c_j} = \lambda$ and that the total attention is fixed and can just be reallocated $\sum_j \frac{\partial t_j}{\partial c_i} = 0$, the third term is zero. Since $u_i(q_i) > 0$ by assumption, we can rewrite the above as

$$\frac{\partial v_i \left(c_i, \frac{t_i}{c_i} \right)}{\partial c_i} = \frac{\partial v_i \left(c_i, \frac{t_i}{c_i} \right)}{\partial d_i} \frac{1}{c_i} \frac{t_i}{c_i}, \tag{B14}$$

which is equivalent to the first-order condition for complexity choice (13) in the baseline model with all θ's being equal, once we substitute out the shadow cost of attention λ using the first-order condition for attention allocation (5).

Comparing the first-order conditions for complexity choice in the competitive equilibrium (B9) with that in the social optimum (B14), the difference between the two is $\frac{\partial t_i}{\partial c_i}$, exactly as in the baseline model, i.e., the difference between (9) and (13). Therefore, the results on the distortion of complexity derived

in the paper extend to this more general setting. Comparing the competitive first-order condition for quantity choice (B7) with the social optimum (B12) it is clear that there are additional inefficiencies arising from monopolistic supply restrictions. However, these effects are orthogonal to the externalities in complexity choice that we focus on in this paper.

Intuition. The above analysis shows that with continuous quantity choice, prices alone cannot correct complexity inefficiencies. This contrasts with the case of unit goods, where setting prices before consumers allocate attention can resolve the complexity inefficiency. The reason is that with continuous quantity choice, price is a single instrument that must simultaneously govern both the quantity consumed and the complexity chosen, and therefore cannot efficiently address both margins. For unit goods, prices do not face this dual role and can alleviate complexity inefficiencies, provided the consumer can credibly threaten to forego the good and reallocate attention to other goods (as discussed in Section 4).

Examples. We now provide examples of real-world applications in which the supplier sets both complexity and price, while the consumer makes a continuous quantity choice but must also devote attention to the good. The most natural settings are closed ecosystems—such as technology platforms, gaming systems, or entertainment franchises—where the supplier determines the ecosystem's complexity and the consumer chooses how many goods or services to purchase within that ecosystem. In closed ecosystems, suppliers typically have monopoly power (or at least substantial market power). In technology ecosystems, consumers choose how many devices or add-ons to purchase from a supplier like Apple. In entertainment franchises such as "Lord of the Rings" or the "Marvel Cinematic Universe," consumers choose not only how many films to watch but also how much themed merchandise and experiences to purchase over time.

C More general v_i function

In this Appendix, we show that our results hold for more general v_i functions than the one specified by Assumption 1. We first show that all our results extend to multiplicative (rather than additive) v_i functions. We then show that our results remain qualitatively unchanged if we relax the monotonicity assumption on $\sigma_i(d_i) = -\frac{g_i'}{g_i'' d_i}$ and, in the multiplicative case, also relax the monotonicity of $\frac{f_i'(c_i)\cdot c_i}{f_i(c_i)}$.

C.1 Multiplicative v_i

We introduce a multiplicative v_i function by replacing Assumption 1 with the following.

Assumption 2. Multiplicative v_i:

(i) Assume v_i is multiplicative

$$v_i(c_i, d_i) = f_i(c_i) \cdot g_i(d_i), \tag{C1}$$

(ii) $g_i(d_i)$ is twice continuously differentiable with $g_i'(d_i) > 0$ and $g_i''(d_i) < 0$ for $d_i > 0$,

(iii) $f_i(c_i)$ is twice continuously differentiable with $f_i'(c_i) > 0$ and $f_i''(c_i) < 0$ for $c_i \in (0, c_i^{max})$, and $f_i'(c_i) = 0$ for $c_i \geq c_i^{max}$, where $c_i^{max} \in (0, \infty]$,

(iv) $f_i(c_i) \geq 0$, $g_i(d_i) \geq 0$ and both are bounded from above,

(v) the limits of all above derivatives exist as $c_i \to 0^+$ and $d_i \to 0^+$

(vi) $\sigma_i(d_i) = -\frac{g_i'}{g_i'' d_i}$ is strictly decreasing in d_i,

(vii) $f(0) = 0$ and $\frac{f_i'(c_i) \cdot c_i}{f_i(c_i)}$ strictly decreasing for $c_i < c_i^{max}$.

The only additional assumption required for the multiplicative case is the final part in Assumption 2, which ensures that the attention curve is single-peaked. While the intuition for this condition is not straightforward, it holds for many of the most natural functional forms satisfying Assumption 2; see Footnote 6 for examples. Note that we assumed both that $\frac{f_i'(c_i) \cdot c_i}{f_i(c_i)}$ is strictly decreasing and that $f(0) = 0$, even though the former implies the latter in the limit. The reason we explicitly assumed $f(0) = 0$ is that it has a clear (and plausible) interpretation in the multiplicative case: A good with zero complexity has no value, regardless of the amount of attention paid to it.

In the following, we show that all our main results continue to hold under Assumption 2 (except for Proposition 5, which already refers to a specific additive functional form). Lemma 1 and Propositions 1 to 4 hold verbatim for the multiplicative case and the proofs only rely on items in Assumption 1 that have their exact counterpart in Assumption 2. The counterpart of (14) with multiplicative v_i is

$$\left.\frac{\partial \tilde{t}_i(c_i, \lambda)}{\partial c_i}\right|_\lambda = \frac{g_i'}{g_i''} + \frac{t_i}{c_i} - \frac{f_i' \cdot g_i'}{f_i \cdot g_i''} \cdot c_i. \tag{C2}$$

Recall from Lemma 1 that $\left.\frac{\partial \tilde{t}_i(c_i, \lambda)}{\partial c_i}\right|_\lambda$ has the same sign as $\frac{\partial t_i}{\partial c_i}$. Hence, we have shown the counterpart of Lemma 2 for a multiplicative v_i function:

Lemma 4. *$\frac{\partial t_i}{\partial c_i} > 0$ if and only if $\sigma_i(d_i) \cdot \left(1 - \frac{f_i'(c_i)\cdot c_i}{f_i(c_i)}\right) < 1$ (and vice versa), where $\sigma_i(d_i) \equiv -\frac{g_i'}{g_i'' d_i}$.*

Lemma 3 also holds verbatim but the proof is different, we present it below.

Proof of Lemma 3 for the multiplicative case. Plugging $v_i(c,d) = f_i(c) \cdot g_i(d)$ from Assumption 2 in the first-order condition for attention allocation (5) we get

$$(1-\theta_i) \cdot f_i(c_i) \cdot g_i'(d_i) = \lambda \cdot c_i. \tag{C3}$$

Plugging $v_i(c,d) = f_i(c) \cdot g_i(d)$ into (A19) and using the above for λ we arrive at:

$$\frac{\partial d_i}{\partial c_i} = \frac{f_i'(c_i) \cdot g_i'(d_i) - \frac{1}{c_i} \cdot f_i(c_i) \cdot g_i'(d_i) - \frac{1}{1-\theta_i} \cdot \frac{\partial \lambda}{\partial c_i} \cdot c_i}{-f(c_i) \cdot g_i''(d_i)}. \tag{C4}$$

We know that $f(c_i) \cdot g_i''(d_i) < 0$ by definition. From Lemma 1 we know that $\frac{\partial \lambda}{\partial c_i} > 0$ if $\frac{\partial t_i}{\partial c_i} > 0$ thus a sufficient (but not necessary) condition for $\frac{\partial d_i}{\partial c_i} < 0$ is $f_i'(c_i) \cdot g_i'(d_i) - \frac{1}{c_i} \cdot f_i(c_i) \cdot g_i'(d_i) < 0$, which simplifies to $f_i'(c_i) \cdot c_i < f_i(c_i)$ given that by assumption $g_i'(d_i) > 0$. Since f_i is an increasing concave function, a line with slope $f_i'(c_i)$ through the point $(c_i, f(c_i))$ crosses the vertical axis at $f(c_i) - c_i \cdot f_i'(c_i)$. Because $f_i(0) \geq 0$ by Assumption 2 and f_i is strictly concave, $f(c_i) - c_i \cdot f_i'(c_i) > 0$. Therefore, $f_i'(c_i) \cdot c_i < f_i(c_i)$ holds for all $c_i > 0$. □

Proposition 4 has to be slightly adapted for the multiplicative case. We state it below and provide the part of Proof that is different from the Proof of Proposition 4.

Proposition 6. Complexity and the Depth of Understanding. *Denote the inverse function of the strictly decreasing function $d_i(c_i)$ by $c_i(d_i)$. If the consumer pays any attention to good i ($t_i > 0$), there exists a unique $\bar{d}_i \in (0, \infty)$ defined by $\sigma_i(\bar{d}_i) \cdot \left(1 - \frac{f_i'(c_i(\bar{d}_i))\cdot c_i(\bar{d}_i)}{f_i(c_i(\bar{d}_i))}\right) = 1$ such that the supplier has an incentive to*

(i) overcomplicate goods that are relatively well understood in the social planner's solution $d_i > \bar{d}_i$;

(ii) oversimplify goods that are not well understood in the social planner's solution $d_i < \bar{d}_i$.

Proof of Proposition 6. We show that in this case as well we have $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i} > 0$ for $c_i < \bar{c}_i$ and $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i} < 0$ for $c_i \in (\bar{c}_i, \bar{\bar{c}}_i)$. By Assumption 2, $\sigma_i(d_i)$ is strictly decreasing in d_i and by Lemma 3 d_i is strictly decreasing in c_i, thus we have that $\sigma_i(d_i(c_i))$ is strictly increasing in c_i. Also by Assumption 2, $f' > 0$, $f'' < 0$, and $\frac{f_i'(c_i)\cdot c_i}{f_i(c_i)}$ is strictly decreasing in c_i so $\sigma_i(d_i(c_i)) \cdot \left(1 - \frac{f_i'(c_i)\cdot c_i}{f_i(c_i)}\right)$

is strictly increasing in c_i. Since we have shown that $\lim_{c_i \to 0^+} \frac{\partial t_i}{\partial c_i} > 0$, Lemma 4 implies that $\lim_{c_i \to 0^+} \sigma_i(d_i(c_i)) \cdot \left(1 - \frac{f_i'(c_i) \cdot c_i}{f_i(c_i)}\right) < 1$. In the Proof of Proposition 4, we have shown that there is a $c_i \in (0, \infty)$ at which $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i} < 0$ thus by Lemma 4 at that specific c_i we have $\sigma_i(d_i(c_i)) \cdot \left(1 - \frac{f_i'(c_i) \cdot c_i}{f_i(c_i)}\right) > 1$. Since $\sigma_i(d_i(c_i)) \cdot \left(1 - \frac{f_i'(c_i) \cdot c_i}{f_i(c_i)}\right)$ is strictly increasing in c_i, there is a unique cutoff $\bar{c}_i \in (0, \infty)$ defined by $\sigma_i(d_i(\bar{c}_i)) \cdot \left(1 - \frac{f_i'(\bar{c}_i) \cdot \bar{c}_i}{f_i(\bar{c}_i)}\right) = 1$. From Lemma 4 it then follows that $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i} > 0$ for $c_i < \bar{c}_i$ and $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i} < 0$ for $c_i \in (\bar{c}_i, \bar{\bar{c}}_i)$. Note that we also used the continuity of $\sigma_i(d_i(c_i))$ and $\sigma_i(d_i(c_i)) \cdot \left(1 - \frac{f_i'(c_i) \cdot c_i}{f_i(c_i)}\right)$ in c_i for the above proof which follows from f_i and g_i being continuously differentiable for $c_i > 0$ and also $d_i(c_i)$ being continuous. □

C.2 Relaxing monotonicity assumptions

We now show that our results are qualitatively unchanged even if we do not make any regularity assumptions about the f_i and g_i functions beyond boundedness. In Proposition 7 below, we show that under reasonably mild assumptions for both the additive and multiplicative v_i function, the attention curve has a generalized hump-shape: It is first increasing in c_i and, at some point, converges back to 0 for large c_i. We refer to this as a generalized hump-shape because the derivative $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i}$ could change sign multiple times as c_i increases. The appropriate parts of the proof of Proposition 3 apply to this setting as well.

Proposition 7. General Shape of the Attention Curve. *Assuming only parts $(i) - (v)$ of either Assumption 1 or Assumption 2, the attention curve $t_i(c_i, c_{-i})$ is initially increasing from zero and at some point it becomes decreasing and converges back to zero at $\bar{\bar{c}}_i \leq \infty$. More specifically:*

(i) $\lim_{c_i \to 0^+} t_i(c_i, c_{-i}) = 0$

(ii) $\lim_{c_i \to 0^+} \frac{\partial t_i(c_i, c_{-i})}{\partial c_i} > 0$

(iii) there exists $c_i \in (0, \infty)$, *s.t.* $\frac{\partial t_i(c_i, c_{-i})}{\partial c_i} < 0$

(iv) $\lim_{c_i \to \infty} t_i(c_i, c_{-i}) = 0$

(v) $t_i(c_i, c_{-i}) = 0$ *for* $c_i > \bar{\bar{c}}_i$

(vi) $\bar{\bar{c}}_i < \infty$ *if and only if* $\lim_{d_i \to 0^+} g_i'(d_i) < \infty$

except for the degenerate case in which $t_i(c_i) = 0$ *for all* c_i.

Proposition 7 shows that functions with constant elasticity, such as CES, are incompatible with our framework. The reason is that CES utility violates the boundedness assumptions on the value of goods, which we view as natural in this context. Furthermore, Proposition 7 shows that boundedness alone generates a hump-shaped attention curve (potentially with multiple peaks). The regularity conditions in Assumption 1 ensure this curve has a single peak. Specifically, if $\sigma_i(d_i)$ is strictly decreasing (and, in the multiplicative case, $\frac{f_i'(c_i)\cdot c_i}{f_i(c_i)}$ is also strictly decreasing), then the attention grabbing effect changes sign exactly once, yielding a single-peaked attention curve.